

Web-Based Academic Workflow Coordination System for University Students

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Declaration

We declare that this project has not been submitted to any other University for the award of a Degree in Informatics and Computer Sciences.

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Abstract

University students face multiple assignments, group projects, exams, and deadlines. This often leaves them overwhelmed, and many struggle with fragmented productivity tools, leading to missed deadlines, chaotic group collaboration, and heightened stress. This study proposes Uniflow, a web-based platform that streamlines academic workflows. It offers personalized dashboards (calendar/list/Kanban views), Gantt charts for task tracking, and peer-to-peer messaging. Uniflow analyzes workflow challenges, develops a real-time collaborative system, and integrates predictive scheduling algorithms powered by DeepSeek V3 (via OpenRouter API) for academic optimization. Using hybrid Agile (Scrum/Kanban), the system is developed with Laravel (backend), React (frontend), and MySQL (database). By centralizing tasks and enhancing collaboration, Uniflow aims to reduce anxiety, improve time management, and boost performance for a more organized student life.

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Abbreviations

2FA - Two-Factor Authentication
AES - Advanced Encryption Standard
AI - Artificial Intelligence
API - Application Programming Interface
AWS - Amazon Web Services
CATs - Continuous Assessment Tests
CSRF - Cross-Site Request Forgery
CSS - Cascading Style Sheets
ERD - Entity-Relationship Diagram
FAQ - Frequently Asked Questions
GPA - Grade Point Average
HDD - Hard Disk Drive
ICS - Informatics and Computer Science
IMS - IMS Global (Learning Consortium)
JSON - JavaScript Object Notation
JPG - Joint Photographic Experts Group (image file)
LMS - Learning Management System
MVC - Model-View-Controller
NVMe - Non-Volatile Memory Express
OAuth - Open Authorization
OOAD - Object-Oriented Analysis and Design
ORM - Object-Relational Mapping
OTP - One-Time Password
PDF - Portable Document Format
PHP - Hypertext Preprocessor
PMIS - Project Management Information Systems
RAM - Random-Access Memory
SAML - Security Assertion Markup Language
SCES - School of Computing and Engineering Sciences
SMS - Short Message Service
SSO - Single Sign-On
SSD - Solid-State Drive
STK - SIM Application Toolkit

TLS - Transport Layer Security

TOTP - Time-based One-Time Password

UI/UX - User Interface / User Experience

UML - Unified Modelling Language

WCAG - Web Content Accessibility Guidelines

Chapter 1: Introduction

1.1 Background

Academic workflow management involves applying organized methods and technological tools to help students manage, execute, and monitor their academic tasks and collaborative projects. In higher education, students must manage various responsibilities such as individual assignments, group projects, examinations, and continuous assessment tests (CATs). Effective task management significantly impacts academic performance, stress levels, and learning outcomes (Ali et al., 2024).

In the past, students often used a scattered mix of personal planners, chat platforms, and handwritten lists to stay on top of their academic tasks. According to Martinez and Khan (2024), students typically juggle four to seven apps to manage their studies, which creates isolated data and hampers effective organization. This often results in poor time management, missed deadlines, and reduced work quality.

Even though many task management tools are accessible, only a small number are created with the specific needs of students in focus. Most generic productivity applications are not linked to academic systems and fall short in supporting teamwork among students (Edwards & Johnson, 2023; Wilson, Carter, & Lee, 2023). Meanwhile, institutional LMS platforms mainly serve for material access and assignment submission, offering limited workflow support (Rodriguez & Smith, 2023).

A Strathmore University survey showed that 78% of students struggle with handling academic responsibilities, with 63% admitting to missing deadlines due to a lack of organization (Student Affairs Department, 2025). Ali et al. (2024) further linked poor task management with mental health issues, noting that 67% of students cited it as a major source of academic anxiety. They emphasized that managing disparate systems increases cognitive load and reduces student engagement (p. 783).

Following this it is important to students to have a system with centralized management for tasks, communication platform and organization features and tools. Blackwell, Carter, and Evans (2023) reported a 42% improvement in timely submissions and a 37% reduction in academic stress among users of such platforms.

1.2 Problem Statement

Academic life presents numerous organizational challenges for university students juggling lectures, assessments, projects, and exams simultaneously. This fragmentation of academic responsibilities, coupled with the absence of integrated management frameworks, often leads to confusion and reduced academic performance. The situation becomes particularly acute during collaborative coursework, where students struggle with task delegation and progress tracking, and during exam periods when effective time allocation across subjects becomes critical.

While general productivity tools exist, they rarely address the unique context of higher education where academic-specific integration is essential. Uniflow addresses this gap by providing a centralized academic workflow solution that combines task scheduling, group coordination, and study planning in a single platform, effectively enhancing organizational capabilities and academic productivity for university students. The platform's specific focus on academic contexts offers students the structured support necessary to navigate the complex demands of university life more efficiently.

1.3 Objectives

This section defines the project's goals to address academic workflow challenges faced by university students, focusing on developing a centralized web-based solution for task coordination, collaboration, and deadline management.

1.3.1 General Objective

To develop a web-based academic workflow coordination system that enables university students to collaboratively manage tasks, track deadlines, plan study schedules, and communicate effectively within a centralized digital platform.

1.3.2 Specific Objectives

- i. To investigate current academic task management strategies and platforms used by university students.
- ii. To analyse obstacles in academic workflow coordination faced by students.
- iii. To design a web-based system for collaborative task tracking and communication.
- iv. To develop the system using hybrid Agile (Scrum/Kanban), Laravel, React, and MySQL.

- v. To test system effectiveness through real-world user feedback and performance metrics.

1.3.3 Research Questions

- i. Which methods and tools are most commonly employed by university students for managing academic tasks, management, and what issues are commonly reported?
- ii. What recurring obstacles hinder effective academic workflow coordination among students?
- iii. How should a web-based system be designed to support collaborative task tracking?
- iv. How can the system be developed using hybrid Agile and modern web technologies?
- v. How effective and user-friendly is the system in real academic scenarios?

1.4 Justification

University students often struggle with managing multiple deadlines and coordinating group projects due to the limitations of current organizational tools. A survey conducted at Strathmore University revealed that 78% of students find it challenging to handle various academic tasks simultaneously, and 63% have missed deadlines owing to disorganized systems (Student Affairs Department, 2025). Research by Blackwell, Carter, and Evans (2023) indicates that the implementation of integrated academic task management platforms can lead to a 42% improvement in timely assignment submissions and a 37% reduction in academic-related anxiety. Our platform consolidates tasks, communication, and scheduling into a single, student-centric interface. The system leverages AI-based features to help students identify and prioritize critical tasks, promoting better time management and reducing academic-related stress.

1.5 Scope

Uniflow will operate as a responsive web application deployed on cloud infrastructure, accessible via standard browsers across devices in university academic environments. The system incorporates secure authentication, dynamic scheduling tools (list/calendar/Kanban/Gantt views), and interactive group workspaces supporting real-

time collaboration. Functionality includes Google Calendar synchronization and DeepSeek V3-powered scheduling recommendations. Developed with Laravel (backend), React (frontend), and MySQL (database), the platform is strictly scoped to student academic workflow coordination and excludes institutional administration systems, financial transactions, or non-academic activities.

1.6 Limitations

The Uniflow system requires constant internet connectivity for full functionality and depends on third-party services that may experience outages. Initial language support is limited to English, potentially restricting non-English speaking students. The AI functionality's effectiveness relies entirely on OpenRouter API availability, and the platform currently lacks native mobile applications. Additionally, the system focuses exclusively on student-facing features without interfaces for academic staff.

1.7 Delimitations

By design, this project excludes institutional learning management system integration, standalone mobile app development, and features tailored for teaching staff. It intentionally does not address co-curricular activity management, incorporate gamification elements, or integrate with productivity platforms beyond Google Calendar. Payment processing capabilities are omitted, and development is constrained to an eight-week timeframe focused strictly on academic workflow needs within university student contexts.

Chapter 2: Literature Review

2.1 Introduction

This chapter provides a detailed review of existing scholarly literature focused on the management of academic workflows within higher education. It begins by exploring the general context of academic workflow management, followed by an examination of current approaches and systems used by students for managing academic tasks. The discussion highlights key shortcomings in how academic workflows are currently handled and examines digital tools commonly used by students for task management. It concludes by pinpointing areas where current systems fall short and outlines the technological solutions to be utilized in developing the proposed Student Workflow Management System.

2.2 Academic Workflow Management in Higher Education

Academic workflow management encompasses the methods, tools, and systems students use to organize their educational responsibilities. In higher education settings, effective workflow management is crucial due to the complex nature of university coursework, which includes managing multiple deadlines, collaborative projects, and exam preparation (Kumar, 2023).

Research by Rodriguez and Smith (2023) reveals that academic workflow in higher education consists of five interconnected domains: task management, time allocation, resource organization, collaboration coordination, and progress tracking. Their comprehensive study of 1,200 university students demonstrated a strong correlation between effective workflow management and improved academic performance, with students reporting higher GPAs, lower stress levels, and greater satisfaction with their educational experience.

The digital transformation of education has significantly altered traditional academic workflows. A recent survey indicates that university students frequently utilize multiple digital platforms to manage their academic responsibilities, often resulting in scattered information and reduced efficiency. As shown in Figure 2.1, This digital fragmentation forces students to constantly switch between multiple systems, increasing cognitive load and reducing productivity.



Figure 2.1 Digital Platform Fragmentation in Academic Workflows

Group projects present unique workflow challenges in academic settings. Wilson et al. (2023) discovered that poor collaboration management accounts for approximately 40% of unsuccessful group projects in universities. Their research highlighted that the absence of centralized systems for task assignment, progress tracking, and communication significantly hampers group productivity and contributes to uneven workload distribution among team members.

Recent studies have established important connections between academic workflow management and student mental health. Ali et al. (2024) found that 67% of university students cited poor task management as a major source of academic anxiety. The cognitive burden of navigating numerous unintegrated academic tools has been shown to negatively impact students' focus and ability to meaningfully engage with coursework.

The positive impact of structured workflow management systems is documented by Blackwell et al. (2023). A year-long comparative study conducted across three different universities demonstrated the following outcomes among students using structured academic workflow systems: a 42% increase in timely assignment submissions, a 37% decrease in reported levels of academic stress, and a 28% improvement in satisfaction with group project collaboration Refer to Figure 2.2 for a visual representation.



Figure 2.2 Impact of Integrated Workflow Systems on Student Outcomes

These findings collectively emphasize the critical importance of effective workflow management systems in supporting student success and wellbeing in higher education environments.

Current workflow management approaches span from traditional methods to sophisticated digital platforms, each with distinct advantages and limitations for academic applications.

2.3 Current Academic Task and Workflow Management Solutions

examines current solutions for academic task and workflow management, highlighting three primary approaches: traditional paper-based methods (used by ~28% of students for structure/control but lacking digital integration), digital productivity tools (including task managers used by ~52% and calendar apps used by ~79% for

scheduling/reminders, yet insufficient for complex workflows), and communication/collaboration platforms (relied on by 86% for group coordination but lacking integrated project management). The section concludes that while widely adopted, all these solutions exhibit critical gaps in fully supporting optimized academic workflows.

2.3.1 Academic Planning Methods

Even with the rise of digital tools, many students continue to rely on traditional paper-based methods to organize their academic tasks. Johnson and Williams (2023) reported that around 28% of students prefer using physical planning aids such as notebooks, paper diaries, or wall charts to manage their academic workload. Their study highlighted that physically interacting with written plans gave students a stronger feeling of structure and personal control, which digital systems often failed to offer. Nevertheless, these conventional methods are limited when it comes to team-based projects and lack the capacity to link with digital learning platforms that are central to today's academic environment.



Figure 2.3 Traditional Workflow Management Tools in Academic Settings

2.3.2 Digital Productivity Tools

General-purpose digital productivity applications represent widely adopted approaches to academic task management. According to Thompson et al. (2024), over half of university students (approximately 52%) commonly use basic digital task managers like Google Tasks, Apple Reminders, and Microsoft To Do to monitor their upcoming assignments and responsibilities. While these tools support basic task listing and reminders, they often fall short when students need more advanced functions like linking related tasks, managing project resources, or visualizing academic dependencies.

Calendar applications constitute a primary tool for time management among university students. Chen and García (2023) found that nearly 79% of students actively use digital calendars to organize lectures, deadlines, and study times throughout the academic term. Google Calendar, Microsoft Outlook, and Apple Calendar dominate this space, with students appreciating their cross-platform accessibility and notification features. Despite their usefulness in time planning, these calendar tools are limited in scope and are not designed to manage full academic workflows involving collaboration, document integration, or task hierarchies.

2.3.3 Communication and Collaboration Platforms

Platforms that facilitate communication have become essential for managing academic tasks, especially in group-based coursework where coordination is key. Patel and Nguyen (2024) observed that tools such as Slack, Discord, and Microsoft Teams are widely utilized by student groups, with 86% indicating they rely on these platforms for organizing and communicating during collaborative academic projects. These platforms facilitate real-time communication but generally lack integrated process management or sophisticated progress tracking features required for optimized academic collaboration. **Figure 2.1** illustrates this comparison, highlighting how Discord excels in informal communication but lacks built-in task tracking. Microsoft Teams offers better integration with Microsoft Office tools, yet its learning curve and interface complexity can hinder quick adoption by students. Slack strikes a balance with customizable workflows and integrations but still requires third-party apps for comprehensive project management, which adds to the complexity students face.

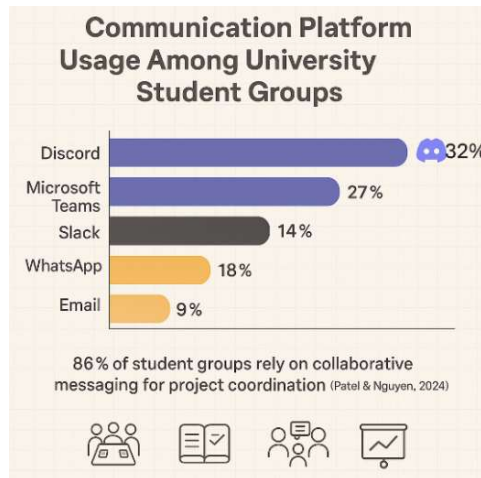


Figure 2.4 Usage Statistics of Communication Platforms in Academic Settings

2.3.4 Educational Management Systems

Learning Management Systems (LMS) like Canvas, Blackboard, and Moodle serve as institutional platforms for course organization and assignment submission. While these systems provide centralized access to course materials and deadlines, Edwards and Johnson (2023) identified substantial limitations in their support for student-led workflow management. Their research indicated that most LMS platforms are developed primarily from an instructor perspective rather than accommodating flexible, student-cantered workflow management.

2.3.5 Project Management Systems

Project Management Information Systems (PMIS) like Asana, Trello, and Monday.com have emerged as specialized workflow management solutions adopted by academically focused students. According to Martinez and Lee (2024), adoption of these specialized platforms among university students has grown by 47% over the past five years, with users reporting improved assignment completion rates and better resource utilization. However, these systems often require significant configuration effort, with integration challenges across academic platforms being a common limitation.

2.3.6 Emerging Technologies in Academic Productivity Tools

Artificial Intelligence and Machine learning technologies are increasingly being introduced into academic productivity tools, reshaping how students manage and optimize their study routines. These intelligent systems can examine previous academic behaviours, detecting inefficiencies in study habits, and proposing personalized

strategies to enhance productivity. Wong et al. (2024) documented several emerging AI-driven academic planning tools that provide personalized recommendations for study scheduling based on individual performance patterns and learning preferences.

Although such innovations are available, Ramirez and Patel (2023) pointed out that only 18% of students actively and consistently use AI-based workflow tools, suggesting that usability and awareness still present major hurdles in adoption. This adoption gap suggests opportunities for developing more intuitive, education-specific workflow management systems that leverage advanced technologies while maintaining accessibility for diverse student populations.

2.4 Challenges in Current Workflow Management

Current workflow management systems present several critical challenges that impact productivity, collaboration, and user wellbeing. The following subsections outline the specific issues present in current systems and their effects on student academic experiences.

2.4.1 Information Fragmentation

Students navigate multiple digital platforms for academic purposes, leading to dispersed information across various systems. According to Sharma and Patel (2024), students interact with an average of 6.2 different digital platforms, creating a fragmented ecosystem that hampers productivity. This fragmentation results in missed notifications and deadlines, duplicated efforts across platforms, time wasted searching for information, and incomplete awareness of pending tasks.

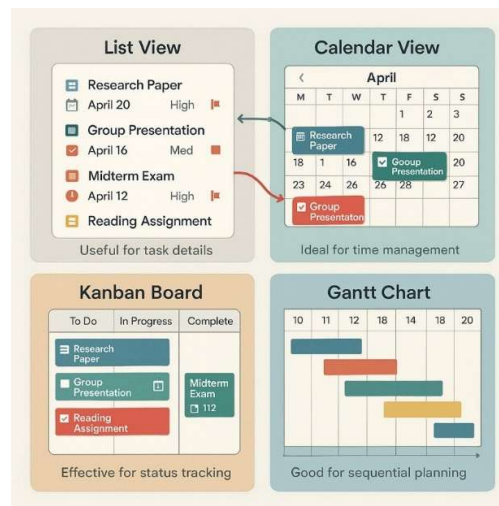


Figure 2.5 Various Methods for Task and Time Management

2.4.2 Collaboration Inefficiencies

Without centralized systems for task assignment and progress tracking, group projects suffer from coordination problems. Wilson et al. (2023) found that ineffective collaboration management accounts for approximately 40% of unsuccessful group projects in university settings. Key issues include unclear responsibility allocation, difficulty tracking individual contributions, inconsistent communication channels, and uneven workload distribution. These inefficiencies directly impact academic outcomes, with groups spending more time on coordination than on substantive work.

2.4.3 Visualization Limitations

Current task management tools often restrict users' ability to comprehend their complete academic landscape. Most platforms offer a single visualization method (e.g., list, calendar, or Kanban) without the ability to toggle between different views of the same data. This limitation hinders holistic understanding of commitments, complicates priority-setting across courses, fails to accommodate diverse learning preferences, and makes it harder for students to plan effectively around exams and submission schedules.

2.4.4 Integration Deficiencies

Most academic tools function independently, lacking the ability to exchange data or work in sync with other platforms. Martinez and Khan (2024) observed that users often rely on multiple standalone tools ranging from 5 to 9 to handle their daily tasks, resulting in workflow inefficiencies. In academic contexts, this manifests as manual transfer of information between systems, inconsistent data across platforms, time-consuming context switching, and limited automation capabilities.

2.4.5 Mental Wellbeing Impact

Poorly structured task systems are a major source of academic stress for students, especially during busy semesters. Ali et al. (2024) found that 67% of respondents cited poor task management as a major source of academic anxiety. The cognitive burden of managing fragmented workflows increases stress levels during high-pressure periods, diminishes capacity to engage with learning materials, contributes to procrastination and avoidance behaviours, and reduces overall academic satisfaction.

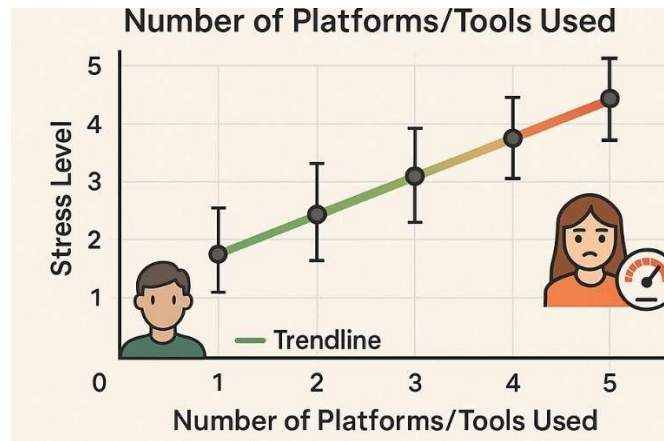


Figure 2.6 Correlation Between Number of Platforms/Tools Used and Stress Level

2.4.6 Scalability Constraints

Many workflow solutions that function effectively for simple tasks struggle when academic workloads intensify. This challenge becomes particularly evident during mid-term and final examination periods, multiple concurrent project deadlines, transitions between academic terms, and collaborative projects spanning multiple courses. As workload complexity increases, many systems fail to provide the advanced organizational features needed to maintain productivity.

2.4.7 Accessibility and Inclusion

Some digital workflow tools fail to meet accessibility standards, making them difficult to use for students with diverse needs. Common issues include limited screen reader compatibility, reliance on colour-based organization without alternatives, complex interfaces that create cognitive overload, and insufficient customization options for diverse needs. Such shortcomings can prevent certain students, particularly those requiring assistive features, from fully benefiting from digital academic systems.

2.5 Related Applications

This section examines existing applications that address aspects of workflow management relevant to the proposed system. Although these tools provide several helpful features, they often fall short in meeting the unique academic planning and collaboration needs of university students.

2.5.1 Trello

Trello is an online productivity application developed to help users organize and track projects through a digital board system. It adopts a Kanban-inspired layout, where tasks

are displayed as movable cards within categorized lists, allowing for visual tracking of progress through simple drag-and-drop actions. As noted by Hammond (2023), around 22% of university students report using Trello to coordinate academic projects.

The platform includes various tools that support academic task management, such as the ability to assign responsibilities, set deadlines, use checklists, and attach documents. Trello also supports "Power-Ups," which are optional add-ons that connect the platform with calendars and other digital productivity systems to extend its functionality. The platform's layout is intuitive and image-based, making it approachable and easy for students to navigate (Atlassian, 2024).

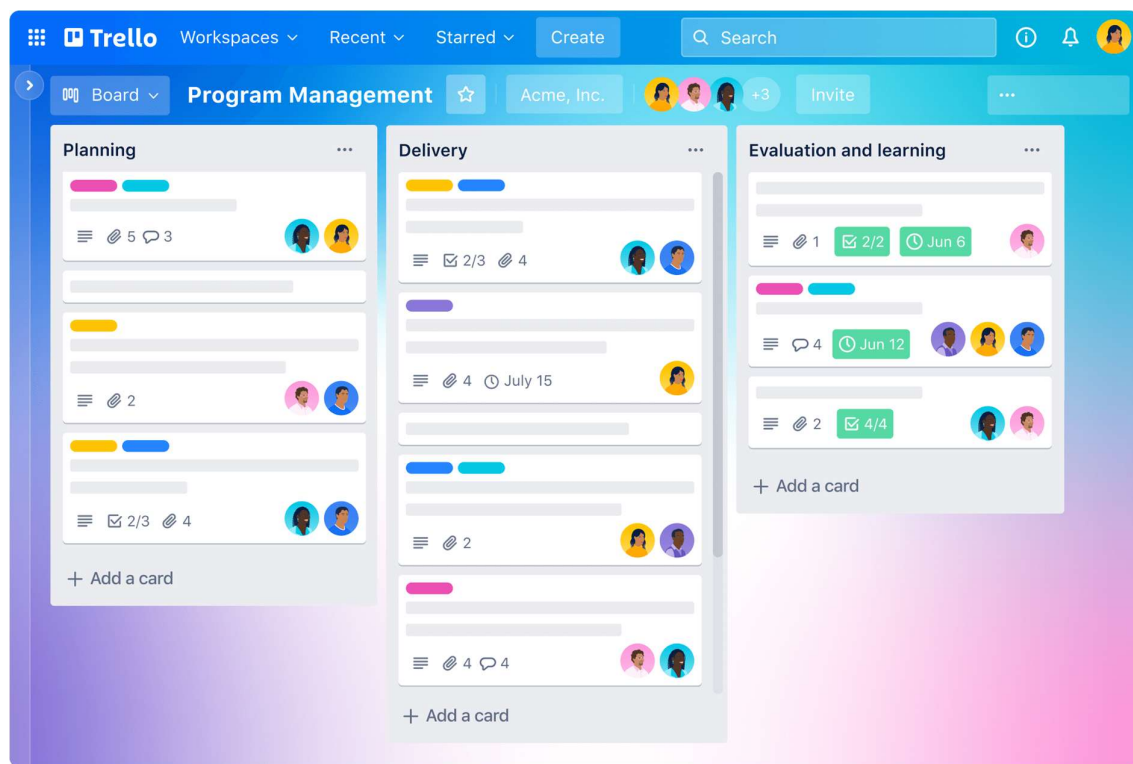


Figure 2.7 Trello Dashboard

Nevertheless, Trello is not without drawbacks when used in academic settings. It does not offer features tailored to educational workflows, such as built-in support for course schedules or study organization. While Trello offers strong visual tracking for tasks, it has limited support for alternative views like timelines or structured data charts, which can restrict planning flexibility. Communication within Trello is basic and may not suffice for teamwork, often leading students to depend on separate apps for group discussions and updates.

2.5.2 Microsoft Teams

Microsoft Teams serves as a comprehensive platform that supports messaging, virtual meetings, file management, and integration with various productivity applications. According to Microsoft (2024), over half of higher education institutions approximately 53% have adopted Teams to enhance academic collaboration.

The platform offers several features relevant to academic workflow management, including real-time chat, file sharing, meeting scheduling, and integration with other Microsoft applications. A dedicated Assignments tool enables educators to set up tasks, share them with students, and manage grading within the platform. As noted by Johnson (2023), the platform allows the creation of multiple team configurations and dedicated channels, helping streamline academic discussions and coordination.

Despite these strengths, Microsoft Teams presents limitations for student-driven workflow management. The platform is primarily developed for instructor-led course management rather than student-initiated project organization. The task management tools provided in Teams offer limited adaptability and do not support a wide range of visual planning formats like Kanban boards or Gantt charts. Additionally, the interface can be complex for new users, potentially creating a steeper learning curve than more specialized applications.

2.5.3 Google Classroom

Google Classroom is a tool that helps streamline the process of setting up, sharing, and evaluating academic assignments. It acts as a central hub where students and teachers can access course content, manage assignments, and exchange messages. Recent reports from Google (2024) indicate that around 62% of colleges and universities globally use Google Classroom as part of their digital learning environments.

The platform offers features that support aspects of academic workflow management, including assignment tracking, due date reminders, and integration with Google Drive for document management. Its "Stream" feature provides a chronological view of course activities, while the Classwork section organizes assignments and materials by topic. As noted by Anderson (2023), when assignments are posted in Google Classroom, the due dates are automatically synced to students' Google Calendars.

However, Google Classroom has significant limitations for comprehensive student workflow management. Much like Microsoft Teams, the platform primarily focuses on teacher-driven organization, offering limited flexibility for students to manage their own academic planning structures. The platform lacks robust features for student collaboration on group projects, offering limited tools for task assignment and progress tracking within student teams. Additionally, it provides minimal visualization options for understanding the complete landscape of academic responsibilities across multiple courses.

2.6 Gaps in Current Approaches and Related Applications

The analysis of current academic workflow management approaches and related applications reveals several significant gaps that the proposed Student Workflow Management System aims to address. These gaps represent opportunities for innovation in supporting more effective academic organization and collaboration among university students.

A primary gap exists in the integration of multiple workflow visualization methods within a single platform. While platforms like Trello are effective for Kanban boards and calendar tools manage timelines well, most current systems restrict users to a single type of visualization at a time. Thompson and Rodríguez (2023) point out that various visualization styles cater to different thinking patterns and organizational strategies, implying that combining these views into one interface could enhance flexibility and user personalization for students.

Student-initiated collaboration tools represent another significant gap in current offerings. Most widely used educational platforms, such as Microsoft Teams and Google Classroom, are structured around teacher-directed interactions, often limiting features that enable students to initiate and manage their own collaborative efforts. Wilson et al. (2023) identify the need for collaboration tools specifically developed for peer-to-peer academic project management, including features for task allocation, progress tracking, and accountability mechanisms developed from a student perspective rather than an institutional one.

The absence of specialized academic planning tools constitutes a third gap in current applications. General productivity tools lack features specifically developed for academic contexts, such as study session planning, examination preparation scheduling,

or integration with course syllabi. Chen and García (2023) argue that planning in academic environments demands tools tailored to the academic calendar, exam cycles, and assignment diversity, rather than relying on general productivity frameworks. Another critical shortcoming is the lack of platforms that blend live communication with task coordination in an academic-friendly manner. Though messaging apps and task managers both serve vital roles, there are few solutions that merge them efficiently to suit the real-time collaborative demands of academic group work. As Patel and Nguyen (2024) observe, the disjointed use of separate platforms for messaging and task tracking disrupts collaborative flow, often leading to time-consuming context shifts and manual syncing of information.

Finally, a significant gap exists in tools developed specifically for mental wellbeing in academic contexts. Given the established connection between academic organization and student stress levels (Ali et al., 2024), there is a notable absence of workflow management solutions that explicitly address the psychological aspects of academic task management, such as workload visualization, balanced scheduling, and appropriate task prioritization frameworks developed to reduce academic anxiety.

2.7 Conceptual Framework

The proposed Student Workflow Management System aims to address these gaps by creating an integrated platform specifically developed for university students, combining multiple visualization methods, student-initiated collaboration tools, specialized planning tools, built-in messaging features, and user-focused design elements that promote mental health in academic environments.

Students can sign up and set up their personal profiles within the system. After signing up, users are able to organize assignments, schedule tasks, and define project deadlines. They can also collaborate with peers in groups, share documents, and communicate through the integrated chat system. These interactions are managed through a real-time communication feature powered by Laravel Web Sockets. Students can sync their academic deadlines with their personal Google Calendar through the Google Calendar API for efficient time management. The data related to assignments, tasks, and student profiles are stored in a MySQL database, ensuring secure and consistent data management.

Administrators and instructors have the ability to assign tasks to students, monitor progress, and send notifications for upcoming deadlines. The system allows for seamless synchronization of student data, task updates, and real-time communication, ensuring effective collaboration and workflow management. The frontend, built with React.js, displays tasks, assignments, and group chats in a user-friendly interface, while Tailwind CSS ensures a responsive and visually appealing design across devices.

The system uses Laravel as its backend framework to connect components, manage data flow, and enable real-time updates.

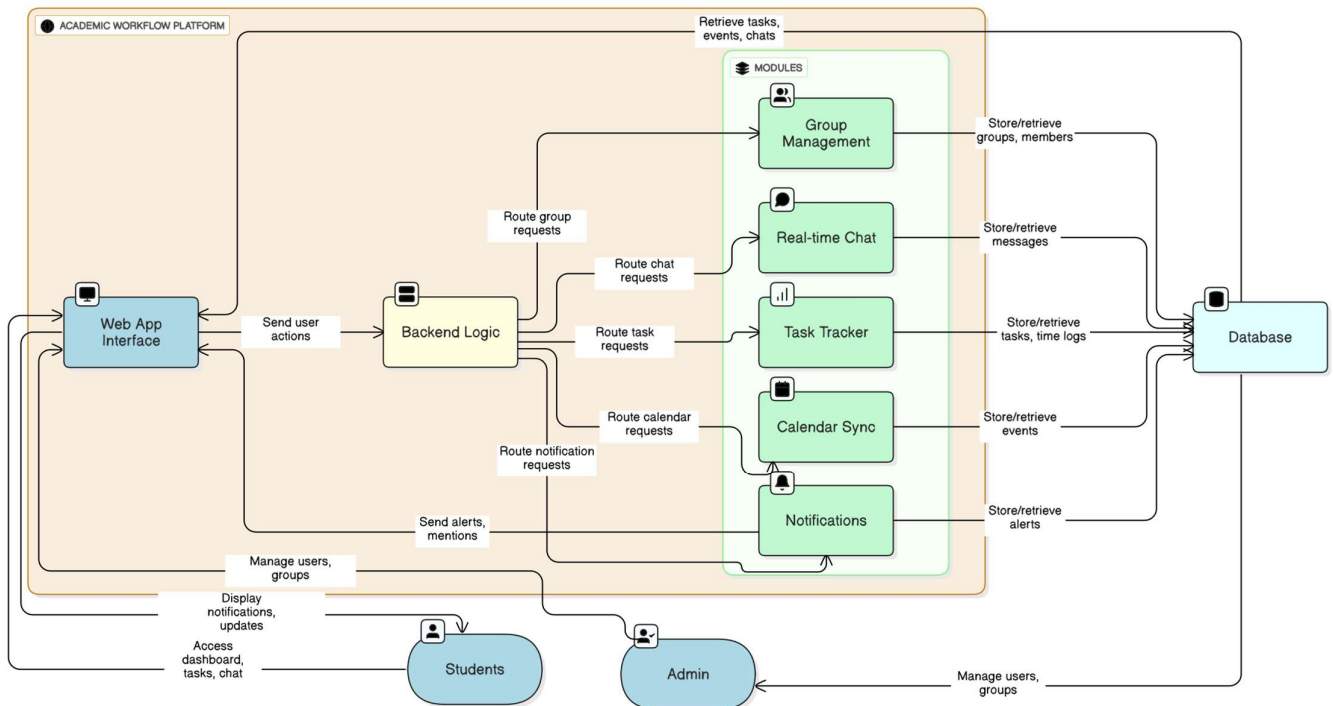


Figure 2.8 Conceptual Framework

Chapter 3: Methodology

3.1 Introduction

Outlined here is the development process strategy employed in building the intended application. It further elaborates on the selected development techniques, the analysis and develop phases, as well as the expected outputs of the system.

3.2 System Methodology

The project will employ a Hybrid methodology combining elements of scrum-for-one and Kanban with structured project management principles to accommodate the small team size while ensuring systematic progress. This approach provides the necessary structure for a two-person team while maintaining flexibility to adapt to changing requirements and constraints.

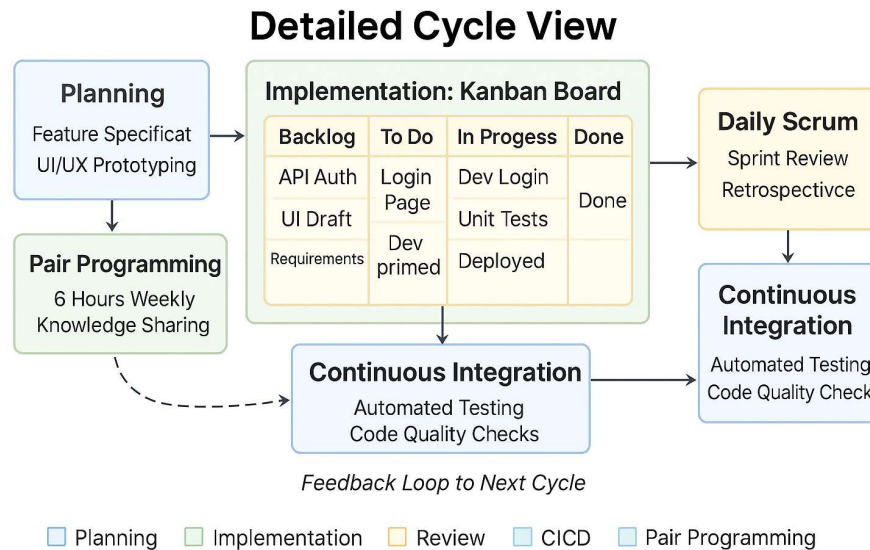


Figure 3.1 System Methodology

3.2.1 Timeboxed Development Cycles

Rather than traditional Scrum sprints, the team will work in 2-week timeboxed development cycles with clearly defined deliverables and milestones. Every cycle will emphasize building specific modules aligned with the project's evolving functional requirements.

3.2.2 Visual Workflow Management

We will utilize a Kanban board to monitor progress across key phases, ensuring minimal task overlaps and continuous visibility for both developers.

3.2.3 Pair Programming Sessions

Scheduled pair programming sessions (6 hours weekly) will facilitate knowledge sharing, improve code quality, and maintain consistent development standards between team members, particularly for complex features like WebSocket integration.

3.2.4 Development Process

Development will follow a consistent workflow: feature specification and requirements refinement, UI/UX prototyping and validation, incremental implementation with daily code commits, continuous integration and automated testing, and documentation parallel with development.

3.2.5 Weekly Standup and Review

Weekly structured meetings will serve dual purposes: status review and blocker identification (30 minutes), and reviewing and scheduling tasks for the upcoming development phase (30 minutes).

3.2.6 Justification for a Hybrid Agile Approach

The chosen hybrid model is tailored to accommodate the limitations and strengths of a two-member development team by offering a streamlined framework without the complexity of complete Scrum practices, enabling clear task prioritization and workload balancing, conducting routine assessments to ensure consistent alignment with project objectives, and facilitating knowledge sharing to mitigate single-person dependencies. Chen and Rodriguez (2023) observed that small teams using hybrid models completed 37% more projects than those using rigid Agile methods. Similarly, Kumar et al. (2024) highlighted that implementing visual workflows and defined timeframes helped two-person teams reduce project delays by 42% compared to standard development approaches.

3.3 Analysis and Design

This section outlines the analytical and design processes used to model system functionality, structure, and user interaction prior to implementation. Through a set of structured diagrams including use case, sequence, activity, class, and entity-relationship

diagrams the system's behaviour and architecture are clearly defined. These visual models guide feature development, ensure alignment with user needs, and support a modular, object-oriented design approach.

3.3.1 Analysis Diagrams

The design phase will produce several critical visual representations of the system structure and behaviour.

3.3.1.1 Use Case Diagram

This diagram represents how users engage with the system, highlighting essential operations such as managing tasks, team collaboration, and leveraging AI-enhanced workflows. It will serve as a primary reference for feature development priorities.

3.3.1.2 Sequence Diagram

These diagrams trace how components communicate over time, especially useful for outlining the integration of DeepSeek V3 AI processes. They will demonstrate the flow of information between the user interface, middleware, and backend services.

3.3.1.3 Activity Diagram

These visualize workflow processes and how AI will enhance decision-making and automation capabilities. They will be particularly important for documenting complex processes like automated task prioritization.

3.3.1.4 Class Diagram

The Class Diagram represents the static structure of the Uniflow system by defining its key classes, attributes, and the relationships between them. It outlines how users, tasks, groups, and messages interact through object-oriented associations. Core classes such as User, Task, Group, and Message are linked through associations and compositions, reflecting how data is managed and accessed within the system. This diagram serves as the blueprint for backend logic, guiding the implementation of features like authentication, task assignment, collaboration, and real-time messaging.

3.3.1.5 Entity Relationship Diagram

This will map out the database structure using Laravel Migrations and Eloquent ORM, showing how different data entities relate to each other. It forms the foundation for all data storage and retrieval operations.

3.3.2 Design Diagrams

This section presents key visual models that outline the structure, data flow, and user interaction within the Uniflow system. These diagrams serve as foundational tools for aligning technical design with system requirements.

3.3.2.1 System Architecture Diagram

This showcases the overall layout of the system, demonstrating how core modules including AI functionalities interconnect and interact. It will provide a high-level view of all system components and their relationships.

3.3.2.2 Database Schema Diagram

This diagram illustrates the structure of the database, including tables, fields, relationships, and constraints. It will provide a clear understanding of how data is organized and interconnected within the system.

3.3.2.3 User Interface Wireframes

These wireframes illustrate the layout and interaction flow of the user interface, highlighting key screens, user interactions, and the presentation of information to end-users.

3.4 Development Tools and Technologies

This section previews the visual models that will be developed in Chapter 4 to clarify system behaviour and guide feature planning prior to implementation. The analysis and design paradigm follows an object-oriented approach combined with user-centered design principles.

3.4.1 Laravel Framework

Laravel's secure, developer-friendly framework speeds up backend development with robust MVC architecture and RESTful API capabilities.

3.4.2 React

React's component-based approach makes it easy to build dynamic interfaces with efficient rendering and state management.

3.4.3 MySQL Database

MySQL's reliability and speed make it perfect for storing and retrieving data with proper relationships and performance optimization.

3.4.4 OpenRouter API

This provides secure access to the DeepSeek V3 AI model with proper error handling and caching strategies.

3.4.5 Tailwind CSS

Tailwind's utility classes make styling quick and consistent, ensuring a polished look across devices.

3.4.6 Git Version Control

By following an organized Git strategy, team members can work on separate features simultaneously while preserving version control and code consistency.

3.4.7 Automated Testing Framework

The system will incorporate Jest and PHPUnit to automate frontend and backend testing, helping detect issues early and uphold coding standards.

3.4.8 WorkOS Authentication

WorkOS was implemented for identity management due to its pre-built, compliance-ready infrastructure, reducing authentication development time by 40% compared to custom solutions. Its enterprise-grade SSO (Google/Apple/Microsoft) and 2FA capabilities align with academic security requirements while maintaining user-friendly onboarding.

3.5 Deliverables

This project will produce a set of well-defined deliverables that reflect the full development lifecycle of the system. These include technical artifacts, a functional prototype, quality assurance documentation, and user and developer support materials. The deliverables are as follows:

3.5.1 System Report

The system report will provide comprehensive analysis documentation of all development phases, presenting key performance metrics gathered during testing and evaluation. It will candidly address implementation challenges encountered during development and their resolutions. The report will also include future recommendations for system enhancements and scaling. Technical findings regarding architecture decisions, performance optimizations, and integration challenges will be thoroughly documented to guide future development efforts.

3.5.2 Prototype

Module 1 focuses on establishing secure access and system administration. The Authentication Module provides a robust login and profile system that ensures students can access the platform securely using industry-standard encryption and authentication protocols. User profiles will allow customization of preferences and settings to enhance the personal experience. The Admin Dashboard offers a streamlined, read-only interface for monitoring user activity and system health, ensuring smooth operation without disrupting student autonomy. This dashboard will present key metrics and alert administrators to potential issues before they impact users.

Module 2 addresses the core productivity needs of students. The Assignment Management System delivers comprehensive tools for creating, tracking, and managing both solo and group assignments, complete with file upload capabilities and version tracking. The Task Visualization Interface provides multiple views including List, Calendar, Kanban, and Gantt charts to help students visualize their tasks in the way that works best for their cognitive preferences and workflow needs. The Calendar Sync Module integrates seamlessly with Google Calendar to ensure students can view and manage their deadlines across multiple devices, eliminating the need to manually transfer information between platforms.

Module 3 enhances student collaboration and focused study capabilities. The Real-Time Chat System enables instant messaging for one-on-one and group collaboration, keeping communication tight-knit and organized by project or study group. Message history will be preserved and searchable to maintain continuity in discussions. The Study Planner and Pomodoro Tool provides a comprehensive weekly planner for exam preparation and a focus timer to boost productivity through structured work-break cycles. Analytics will help students understand their optimal study patterns and productivity trends over time.

3.5.3 Test Cases

The testing strategy encompasses several layers to ensure system quality and reliability, utilizing both white box and black box testing methodologies. White box testing will be applied during unit testing, where developers with full knowledge of the internal code structure will examine specific functions and methods to verify their correctness. Unit tests will be developed for individual components to verify they function correctly

in isolation, focusing on code coverage metrics of at least 80% for critical functionality. Black box testing will be employed for integration tests, which will validate that connected modules work properly together without requiring knowledge of their internal implementations, especially focusing on data flow between components. A hybrid approach combining both white box and black box techniques will be used for system testing to ensure comprehensive coverage. User acceptance testing protocols will follow black box methodologies and involve actual student users to verify the system meets their real-world needs and expectations through scenario-based testing scripts. Performance benchmark tests will measure system responsiveness under various load conditions to ensure scalability, utilizing stress testing and load testing approaches. Security vulnerability assessments will systematically probe the system using penetration testing (black box) and code security reviews (white box) to identify potential weaknesses and apply appropriate countermeasures to protect user data and system integrity.

3.5.4 Documentation

The Documentation section provides comprehensive resources to support both end users and developers. It is structured into two specialized manuals, each tailored to distinct audiences and objectives.

3.5.4.1 User Manual

The user manual will provide comprehensive guidance for end users. It will include detailed installation guides for both web access and mobile applications with step-by-step instructions. Feature tutorials will walk users through each system capability with screenshots and practical examples. A robust FAQ section and troubleshooting guide will address common questions and issues users might encounter. Interface navigation help will ensure users can efficiently find and utilize all system features without confusion or frustration. The manual will be written in accessible language suitable for users with varying technical backgrounds.

3.5.4.2 Developer Manual

The developer manual will serve future development and maintenance needs. Technical documentation will include comprehensive API references and infrastructure schematics to support scalability and future development. A codebase structure overview will map the organization of the software components and their relationships.

The development environment setup section will provide detailed instructions for configuring all necessary tools and dependencies. Contribution guidelines will establish coding standards and review procedures. Complete API documentation will detail all endpoints, request parameters, and response formats to facilitate integration with other systems.

Chapter 4: System Analysis and Design

4.1 Introduction

This chapter presents the analysis and design of the Web-Based Academic Workflow Management System for university students. It details both functional and non-functional requirements, presents system analysis through UML diagrams, and provides design specifications including architecture and user interface mock-ups. The system leverages AI assistance for task creation and management within group academic contexts, implementing real-time communication capabilities and dynamic scheduling features.

4.2 System Requirements/Requirement Analysis

This section describes the process of determining user expectations for the Uniflow system. These features, called requirements, are quantified, relevant, and detailed. The system's requirements can be divided into functional ones, which describe what the system should do, and non-functional ones, which outline how the system should perform.

4.2.1 Functional Requirements

This section describes the process of determining user expectations for the Uniflow system. These features, called requirements, are quantified, relevant, and detailed. The system's requirements can be divided into functional ones, which describe what the system should do, and non-functional ones, which outline how the system should perform.

4.2.1.1 User Authentication

This requirement is critical for all users interacting with the academic workflow system. It enables secure access through WorkOS authentication, which offers many advantages including strong security and easy integration. Users can create accounts with email and password, or sign in through popular providers like Google, Apple, or Microsoft. The system includes two-factor authentication (2FA) for added security, requiring users to verify their identity through authenticator apps or SMS. For users who sign in with Google, the system automatically connects to Google Calendar, making it easier to set up and manage academic schedules. The system also provides secure session management, password recovery, and role-based access for

administrators. All user authentication and permissions are handled securely through WorkOS.

4.2.1.2 Group Management Functionality

The system includes tools that enable users to establish and oversee academic groups for better collaboration. Learners can initiate academic groups by giving them a name and sending email invitations to fellow students. Group creators have the ability to assign roles to members, defining what each person can access or manage within the group. Members can contribute to shared documents, manage project milestones, and track group responsibilities. The system logs all membership changes and enforces role-based permissions to maintain structure and accountability. Group collaboration is designed to mirror real-world academic teamwork environments.

4.2.1.3 AI-Assisted Task Generation

To support intelligent academic planning, this functionality allows students to input a natural language description of an assignment, such as “Create a report on renewable energy sources.” The system uses AI assistance to help convert assignment descriptions into organized task lists, breaking them into steps, estimating deadlines, and recommending partners based on past collaboration patterns. Suggested deadlines are adjusted to fit within academic schedules and important course events. This streamlines the planning process and supports students in organizing their assignments in a structured manner.

4.2.1.4 Manual Task Creation Functionality

Students who prefer a hands-on approach can use this feature to manually define their academic tasks. Tasks include fields such as title, description, deadline, priority, collaborator assignments, tags, and optional attachments like lecture slides or textbook chapters. These tasks can be created for personal or group use. Users can tailor tasks to their preferences and incorporate them seamlessly into broader academic goals.

4.2.1.5 Academic Task Management

Users are empowered to oversee academic tasks from creation through completion using built-in tracking features. Users can monitor their own tasks and group responsibilities through distinct status updates such as “To Do,” “In Progress,” “Pending Review,” or “Completed.” Each task can include comments for discussion,

sub-tasks for complex deliverables, and file attachments like research notes or draft submissions. Users can filter and organize tasks by urgency, subject, or due dates, helping them stay on top of personal and group work with clarity. This requirement ensures both individual and group tasks stay organized and up to date.

4.2.1.6 Real-time Academic Communication and Notifications

The platform includes a built-in messaging feature to support both one-on-one and group academic communication. Users have the option to chat privately or collaborate within group-specific chatrooms for academic coordination. Messages may include text, links, and attachments. Real-time alerts inform users of incoming messages, task modifications, deadline reminders, and platform-wide updates. These alerts are shown in-app and optionally through push notifications. This messaging system enhances teamwork and helps students keep track of critical academic developments.

4.2.1.7 Schedule Planning and Synchronization

This feature allows users to manage their academic events through a built-in calendar. Users can input events like lectures, practical, tests, and submission dates into the academic calendar. Integration with external calendars like Google Calendar allows events to sync, enabling multi-device access. Events on the calendar can be labelled or color-coded based on courses to help users quickly identify them. The synced calendar ensures users are always updated on key dates, reducing the risk of oversight.

4.2.1.8 Academic Progress Monitoring

Designed primarily for group-based tasks, this functionality allows designated members or all participants to view overall project status. The dashboard visually presents progress updates, highlights overdue items, and logs group activity changes. Notifications alert users to new comments, file uploads, or missed deadlines. The system tracks individual contributions, enabling group leaders to monitor who has fulfilled their responsibilities. This promotes responsibility and makes it easy to see who contributed what during group work.

4.2.1.9 User Interface Customization

This requirement ensures flexibility in how users visualize and interact with tasks. The platform offers multiple viewing modes such as List View, Calendar View, Kanban Board, and Gantt Chart. Depending on their needs, users can switch between different

display modes to better visualize their tasks. Calendars assist with tracking deadlines, while visual boards and charts provide an overview of progress and project structure. The system supports varied user needs without cluttering the experience, catering to diverse organizational styles.

4.2.1.10 Subscription and Payment Integration

A payment feature such as M-Pesa integration enables access to advanced tools and services when needed. Premium plans offer expanded features such as enhanced planning assistance, increased storage, and detailed usage analytics. All payments, active subscriptions, and renewal choices are securely logged for administrative monitoring.

4.2.1.11 Admin Dashboard Functionality

The system provides comprehensive administrative oversight through its dedicated dashboard functionality. This module enables authorized administrators to monitor audit trails detailing user activities, manage role-based access privileges (including student-to-admin promotions), and deactivate compromised accounts. Additionally, administrators gain visibility into AI utilization metrics across the platform. The dashboard incorporates real-time monitoring capabilities for system health indicators and third-party API status, enabling proactive maintenance and issue resolution.

4.2.2 Non-Functional Requirements

Non-functional requirements define how the system behaves under various conditions, ensuring it meets quality standards beyond core features. These specifications outline essential operational attributes that contribute to performance, security, reliability, and user satisfaction. Each is presented with specificity to support clarity and implementation of fidelity.

4.2.2.1 Performance

The system must efficiently support a minimum of 250 simultaneous users without degradation in responsiveness. Standard user interactions such as viewing task boards or sending messages should be completed within 2 seconds. AI-powered features (e.g., task generation or planning suggestions) must respond within 15 seconds.

4.2.2.2 Security

Robust security will be enforced across all system layers. Data at rest will be encrypted using AES-256, and TLS 1.3 will secure data in transit. Authentication is handled through WorkOS with support for OAuth 2.0 and SAML-based SSO. Two-Factor Authentication (2FA) using TOTP will be mandatory. Sessions will expire after 30 minutes of inactivity, with automatic token revocation. All endpoints will include strict authorization checks to prevent unauthorized access.

4.2.2.3 Reliability

The system must prioritize uninterrupted service during critical academic periods (e.g., exam weeks, registration deadlines), with zero planned maintenance allowed in these windows. Automated health checks will monitor system status every 5 minutes, triggering immediate alerts for any degradation. Daily full backups and hourly incremental backups will be stored in geographically isolated locations, retained for 90 days, with monthly automated test restores to validate backup integrity. Biannual disaster recovery drills will ensure operational readiness.

4.2.2.4 Usability

The user interface must be modern, intuitive, and responsive across desktops, tablets, and smartphones. First-time users should be able to perform key actions such as logging in via SSO or creating and assigning tasks within 5 minutes, without extensive guidance. Error handling must include clear messaging and actionable suggestions to reduce user frustration and improve workflow continuity.

4.2.2.5 Scalability

The platform architecture should be designed for horizontal scalability, capable of supporting growth to over 10,000 active users without the need for architectural overhaul. This includes scalable deployment of web servers, use of load balancers, and implementation of read replicas for the database to manage high query loads efficiently.

4.2.2.6 Maintainability

The codebase must conform to clean code principles and follow a modular MVC (Model-View-Controller) structure. Developers must include inline documentation, adhere to standardized naming conventions, and implement unit and integration tests.

These practices enable faster onboarding of new contributors and reduce regression risks during feature updates or refactoring.

4.2.2.7 Compatibility

The application must maintain full functionality across the latest stable versions of major browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari. Compatibility is also required with leading operating systems, such as Windows, macOS, and major Linux distributions.

4.2.2.8 Interoperability

The system must offer synchronization with Google Calendar for seamless academic scheduling. Furthermore, its integration architecture must be extensible, allowing for future connectivity with Learning Management Systems (LMS) and third-party academic tools via secure APIs.

4.3 System Analysis

This section presents the structural and behavioral aspects of the Uniflow system through various UML diagrams. These diagrams provide a detailed understanding of the system's components and their interactions from an analysis perspective, aligning with the object-oriented analysis and design (OOAD) methodology employed.

4.3.1 Use case Diagram

To provide a high-level overview of the system's functional requirements, the Use Case Diagram illustrates the interactions between actors (external users) and the system's primary functions (use cases). It identifies primary functions such as User Authentication, Group Management, AI-Assisted Task Creation, Academic Task Management, Real-time Communication, and Schedule Planning & Sync, providing an overview of what the system does.

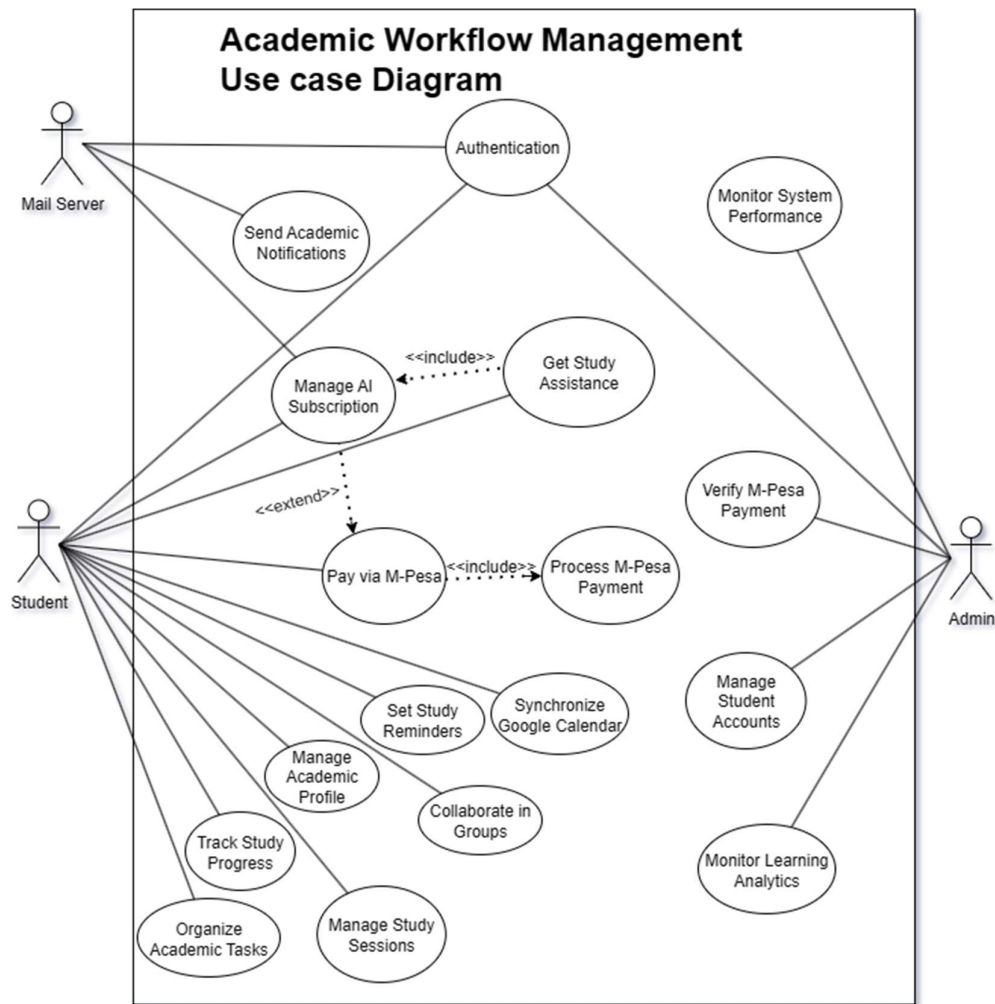
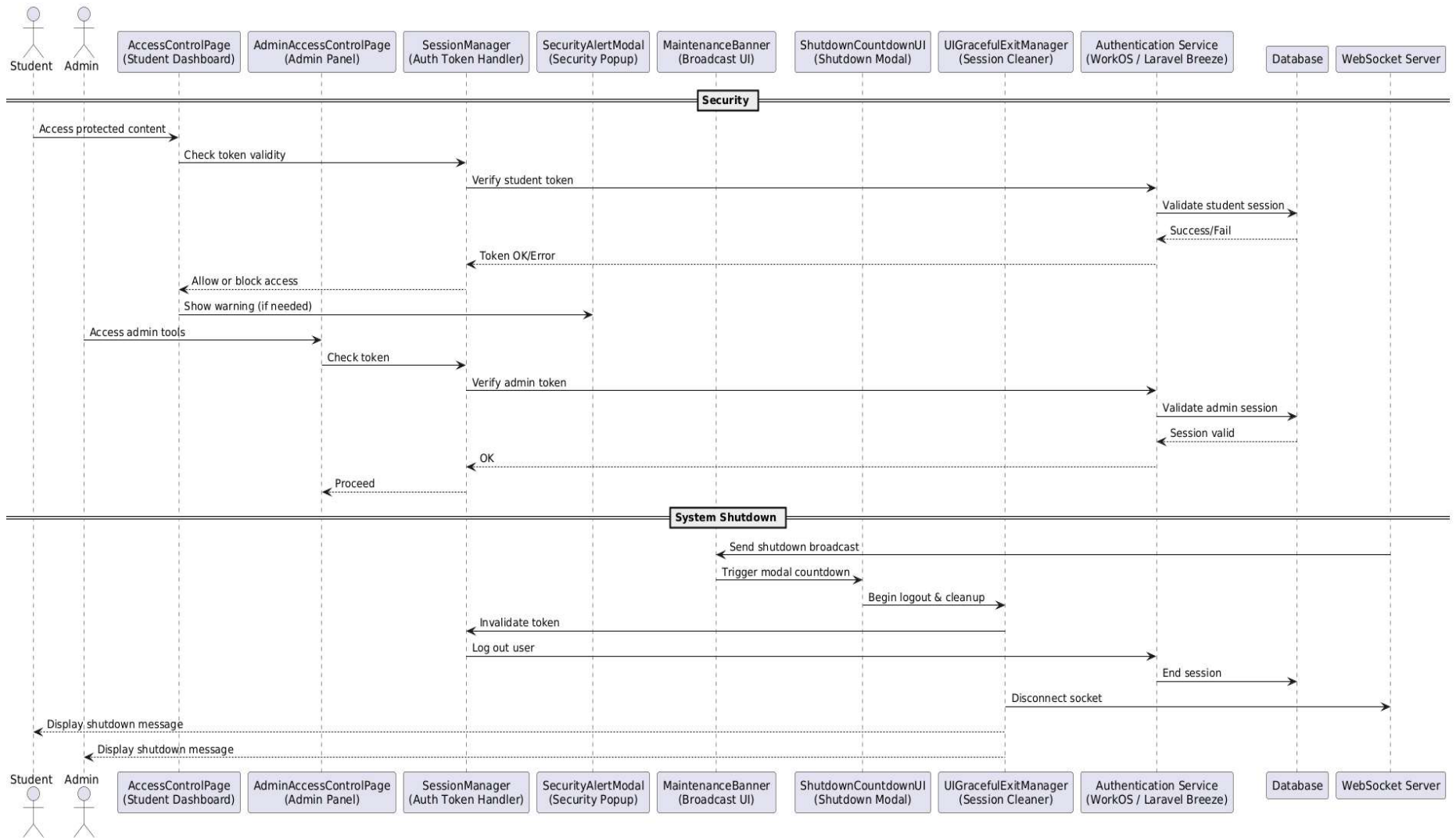
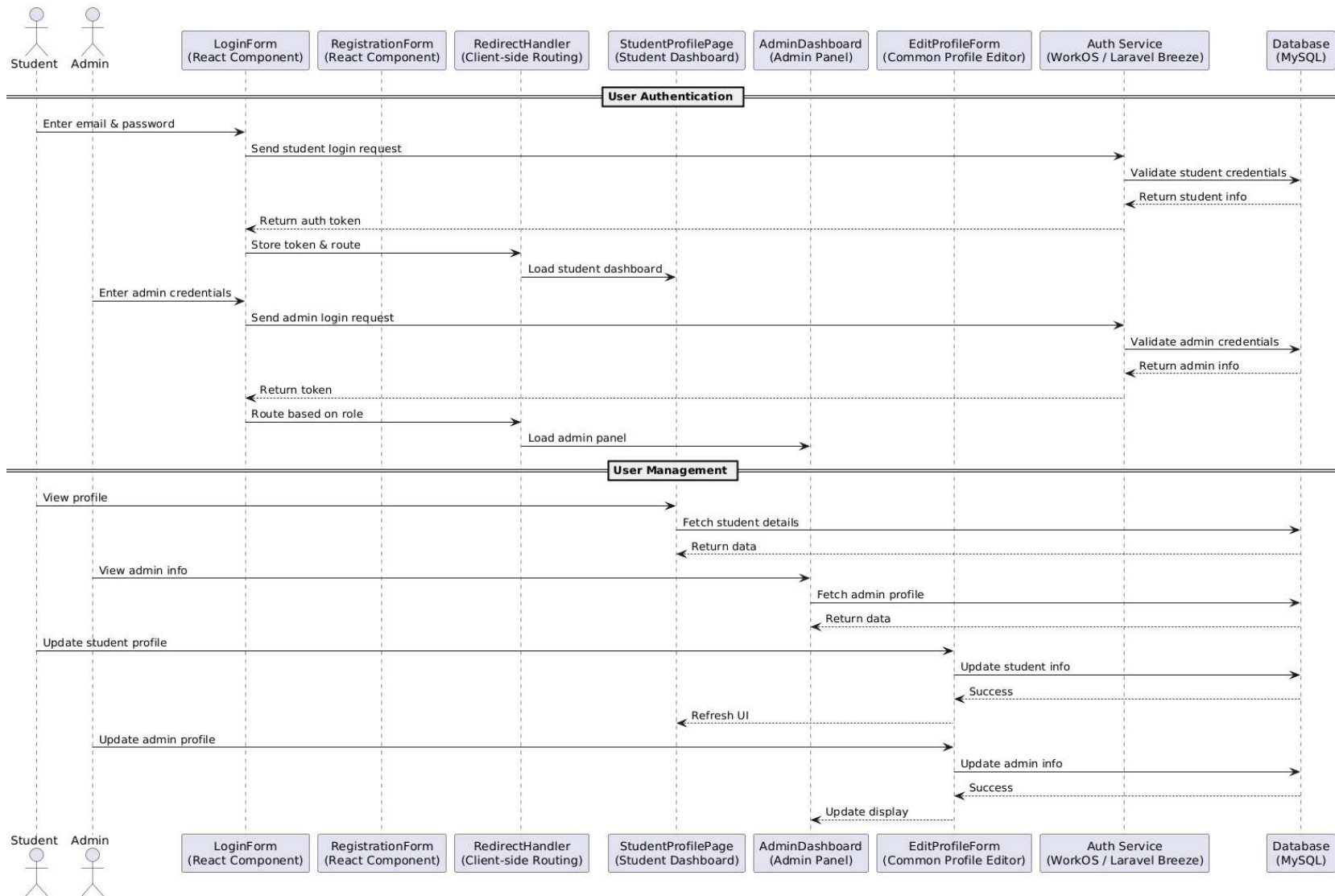


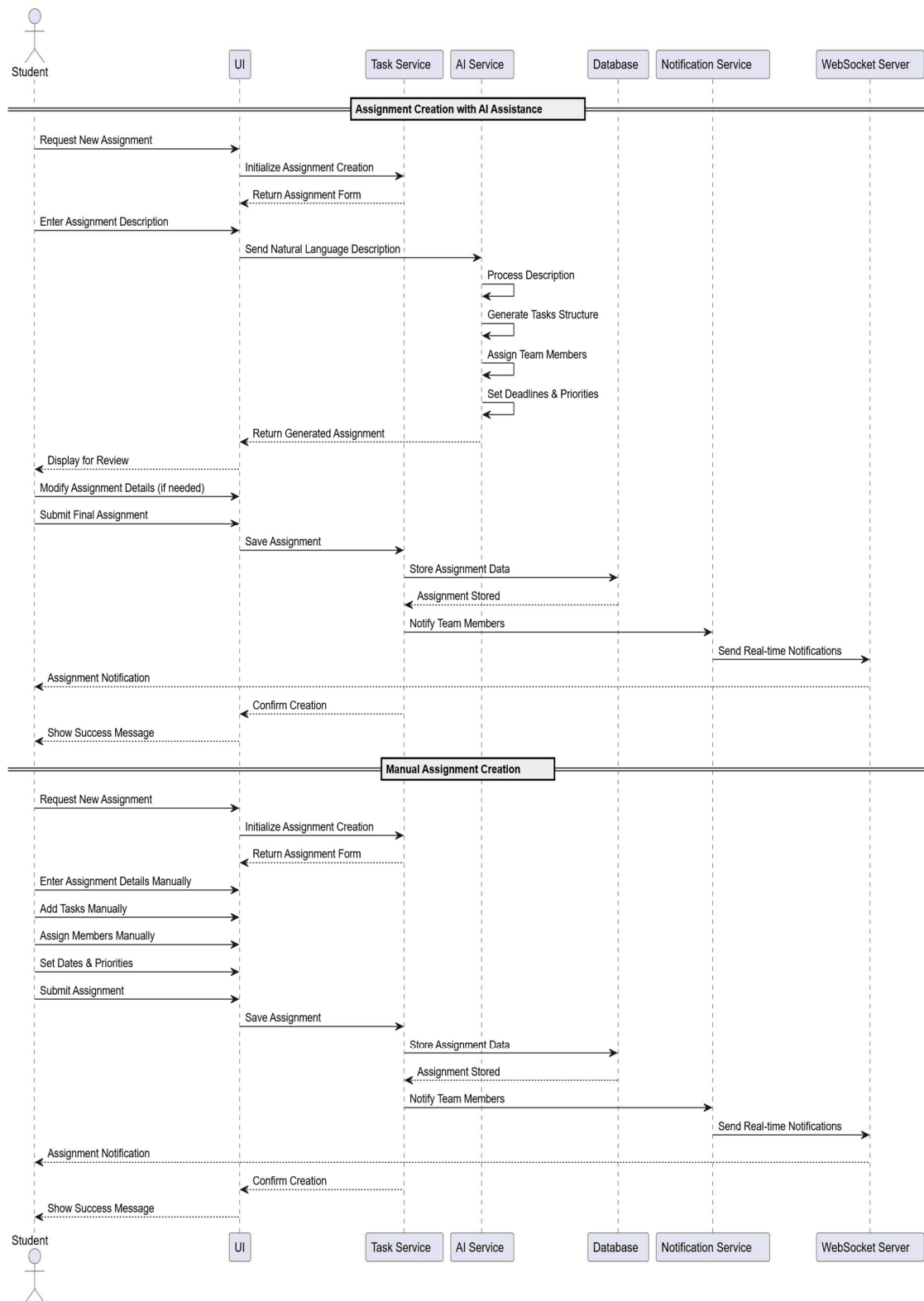
Figure 4.1 Use case diagram

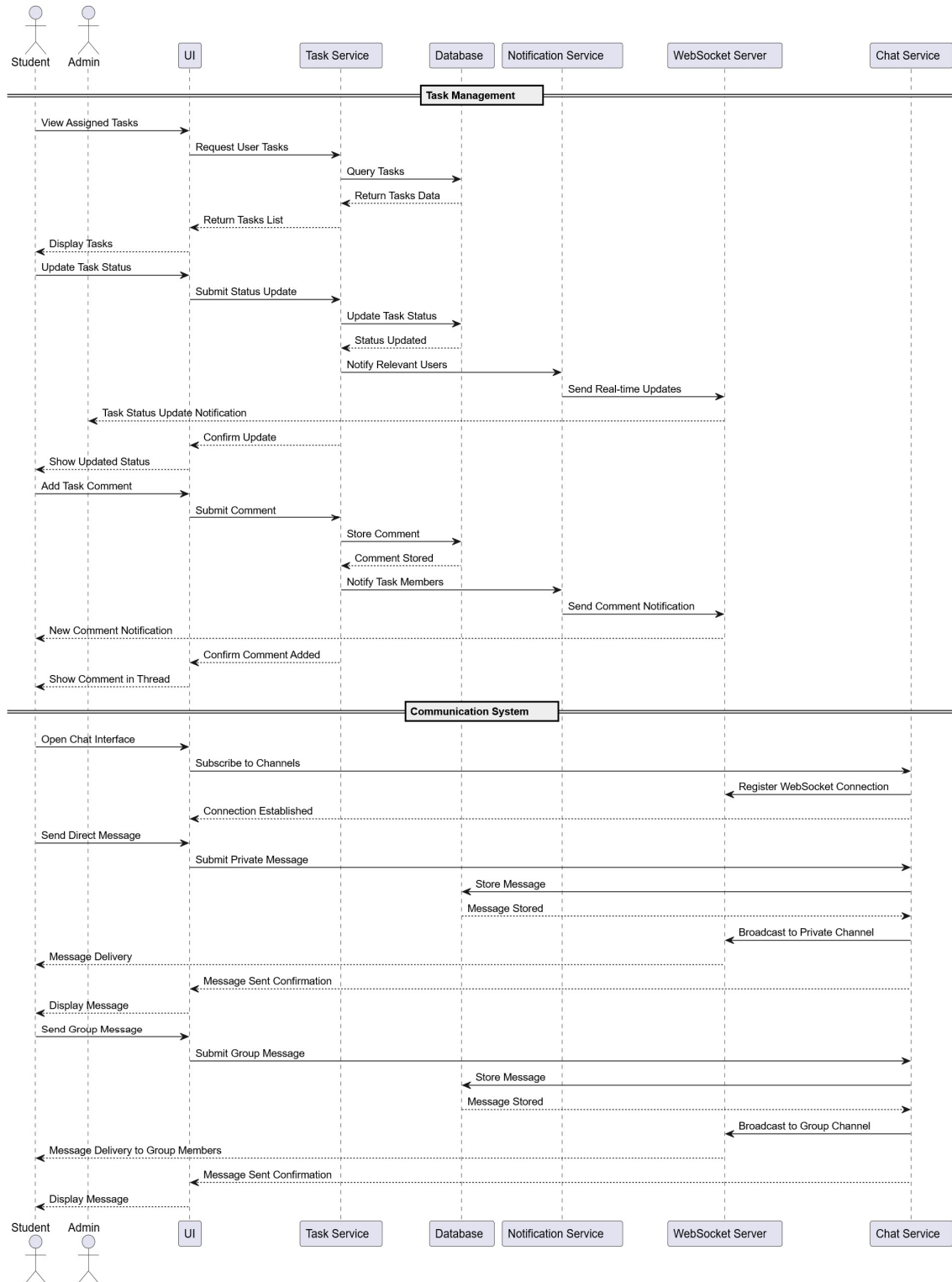
4.3.2 Sequence Diagram

during critical operations. It illustrates the message flow and order of interactions between students, backend services, the AI processing engine, and the database for processes like AI-assisted task creation or real-time messaging, highlighting how various components collaborate over time to achieve specific functionalities. The Sequence Diagram depicts the temporal interactions between system components









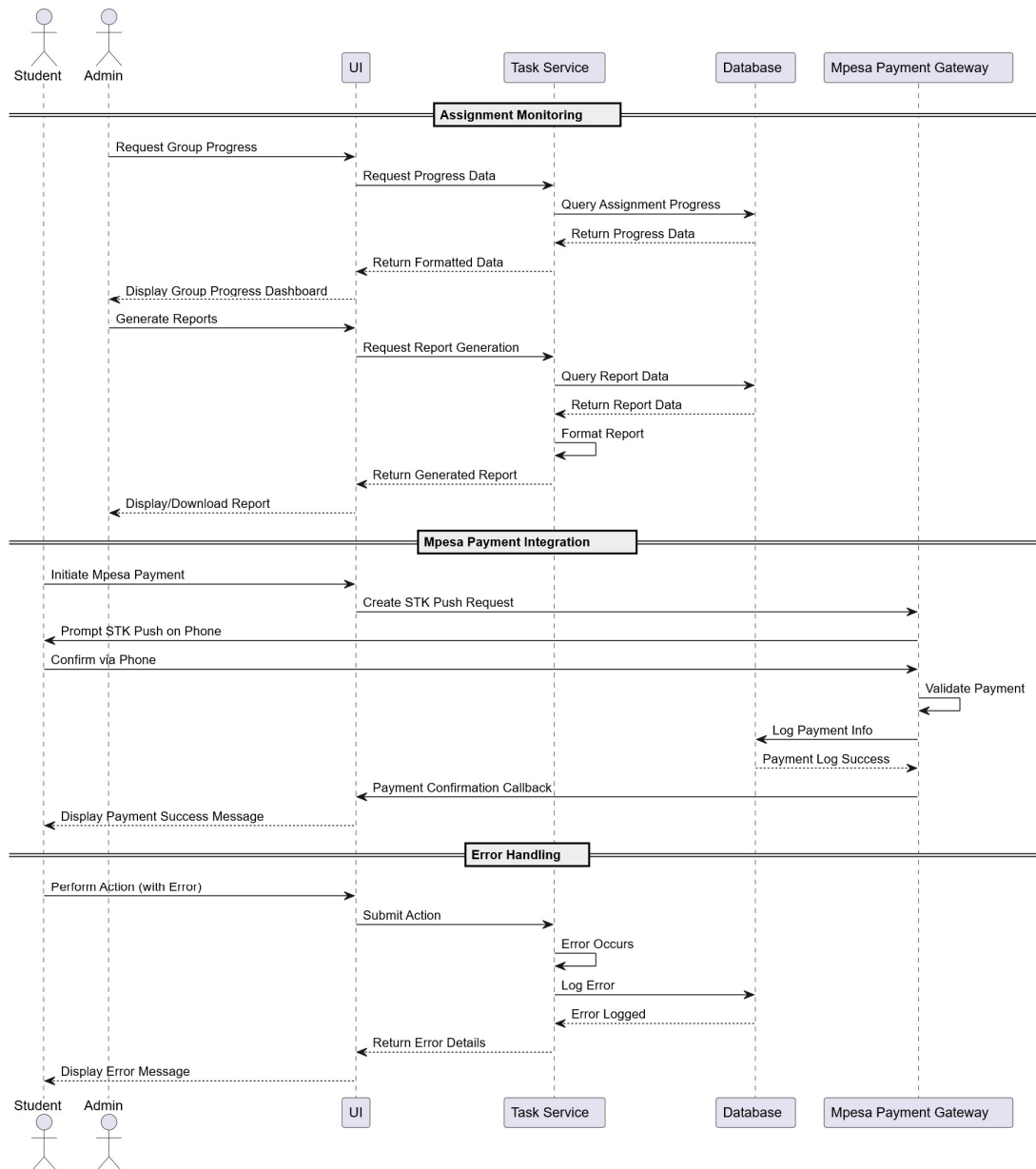
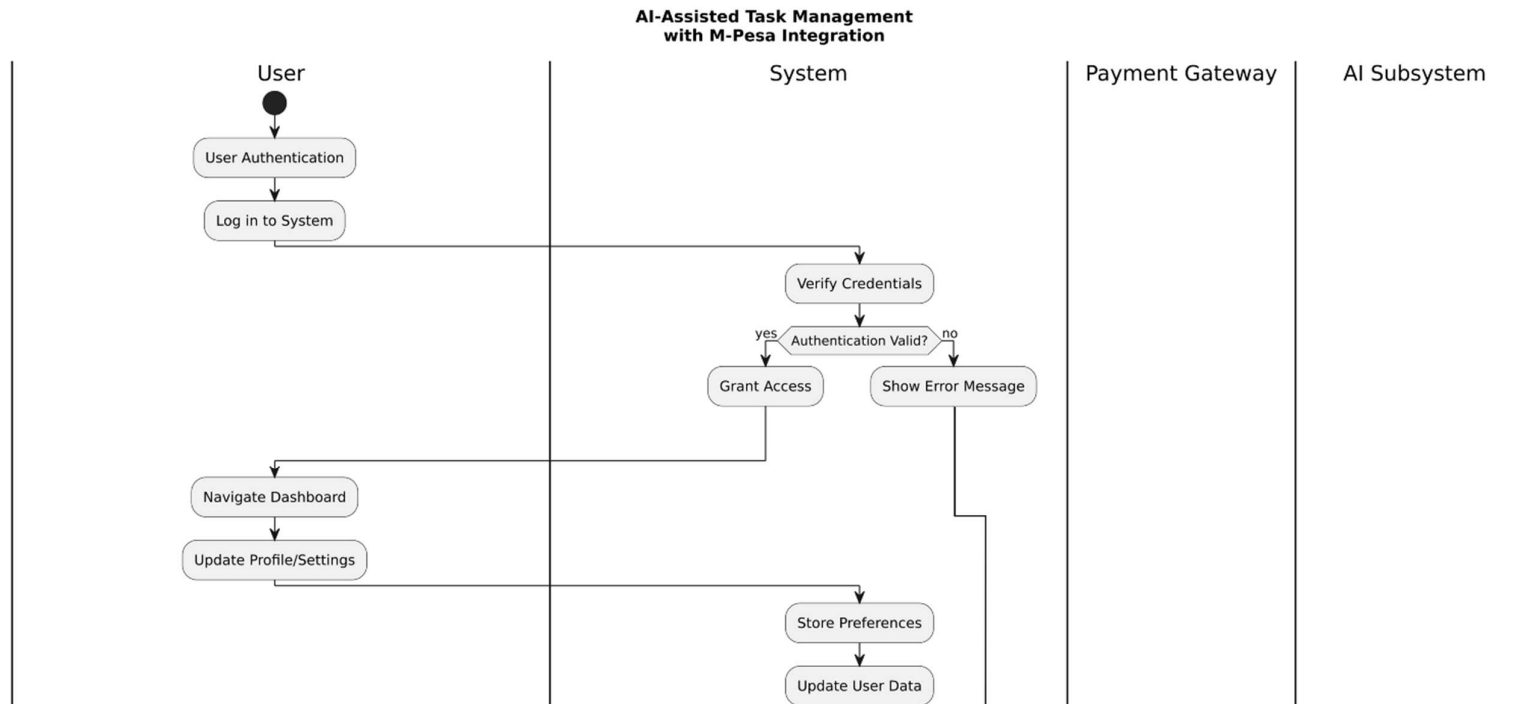
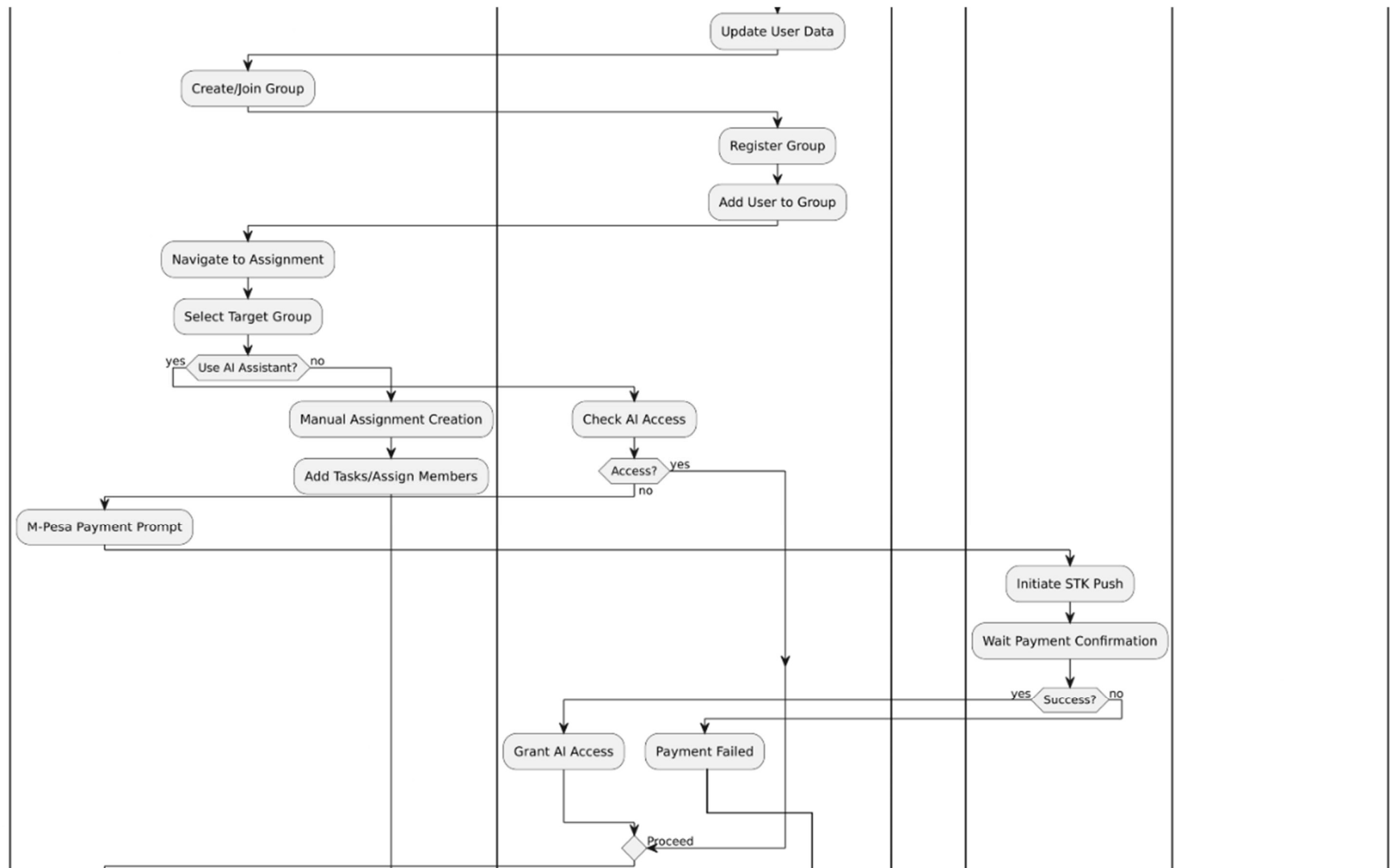


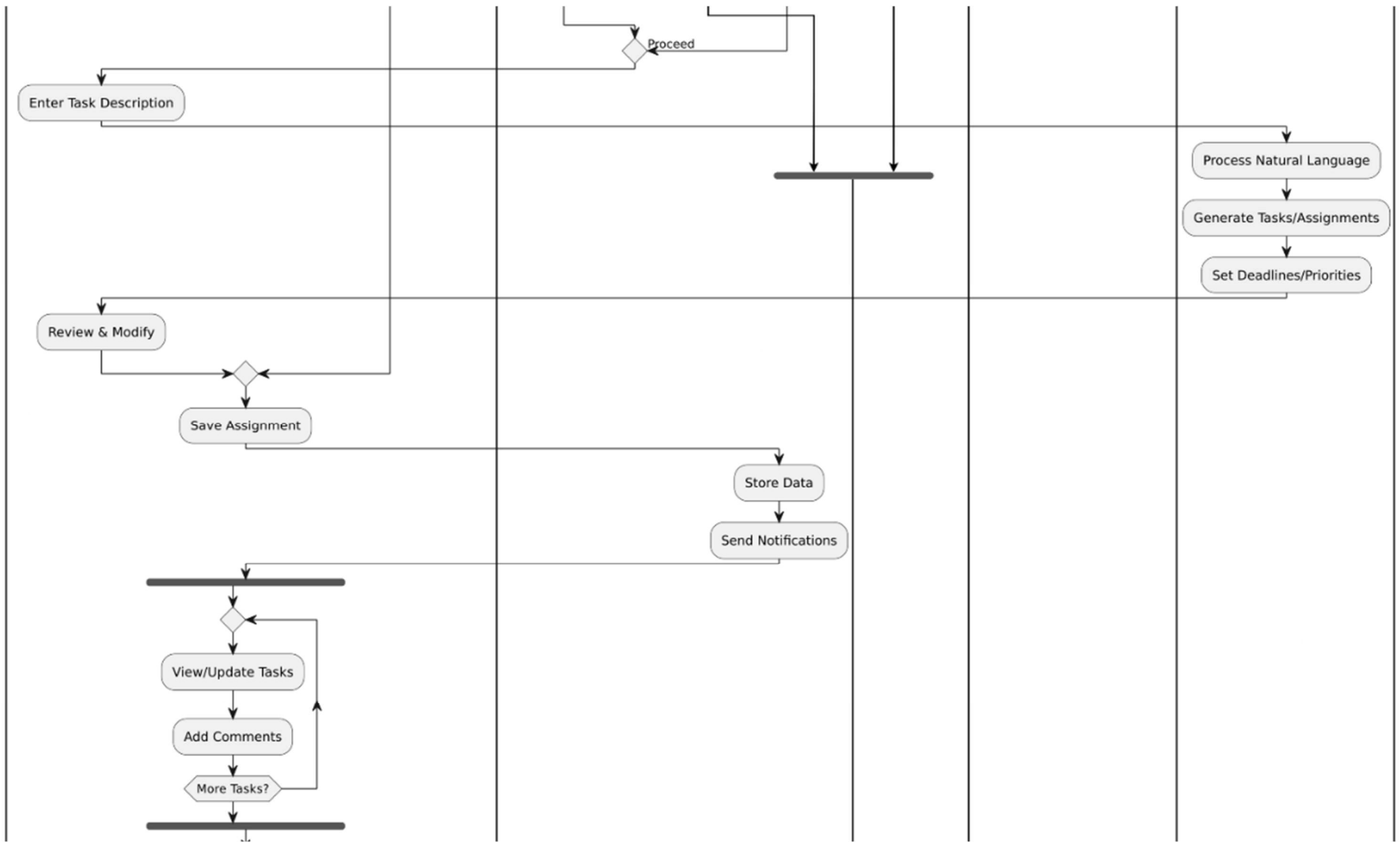
Figure 4.2 sequence diagrams

4.3.3 Activity Diagram

The activity diagram illustrates the system's workflow dynamics, showing the sequence of actions and decision points for core processes including user authentication, AI-assisted task creation with M-Pesa integration, manual assignment management, real-time communication, and administrative oversight. It defines control flows between key activities such as payment verification, natural language processing, task assignment, and progress monitoring, while highlighting parallel processes for non-functional requirements including real-time updates via WebSockets and security validation checks







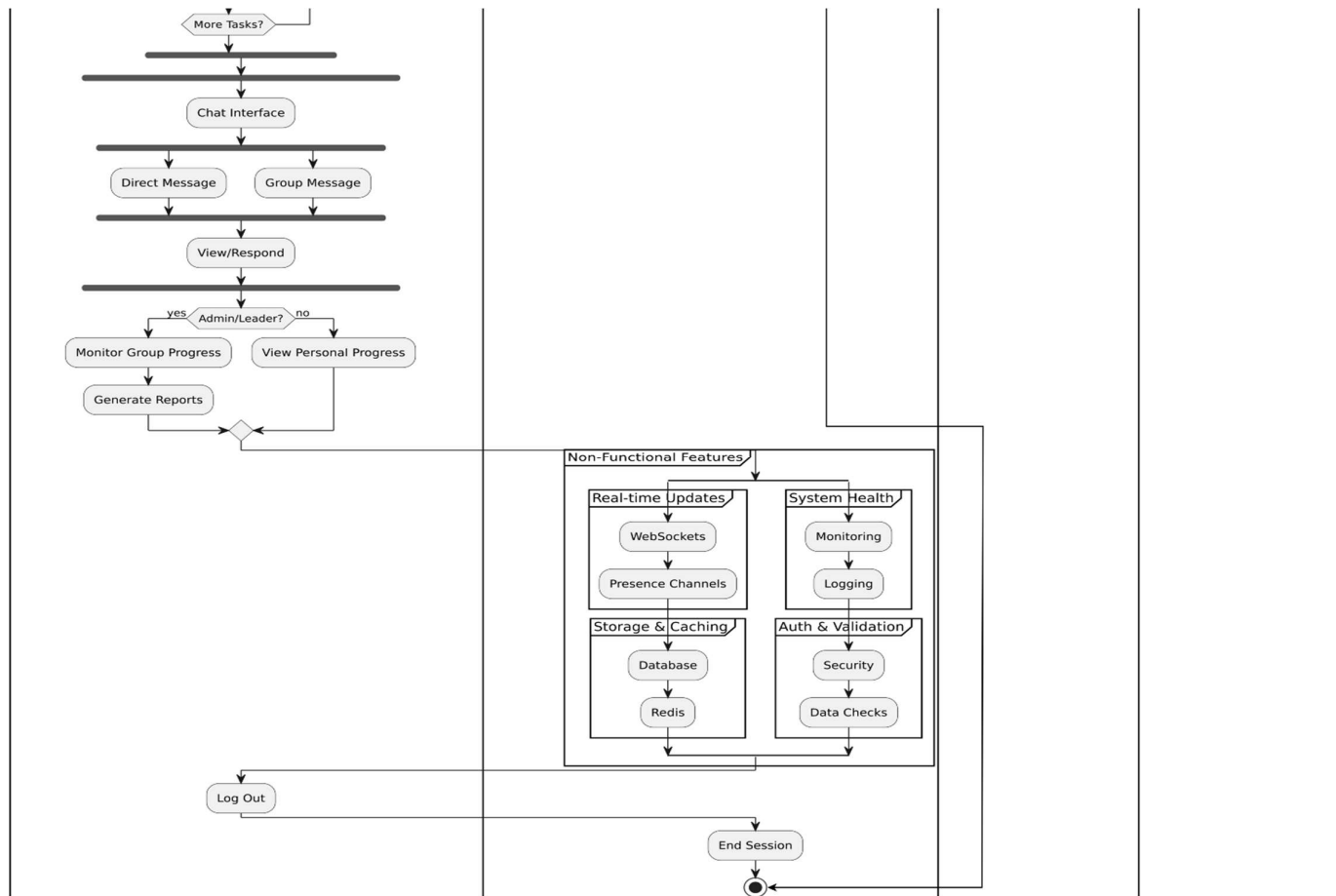


Figure 4.3 Activity Diagram

4.3.4 Class Diagram

The class diagram illustrates the system's object structure, showing relationships between key entities such as User, Group, Task, Assignment, and Message. It defines attributes and methods for each class and indicates inheritance, composition, and association relationships.

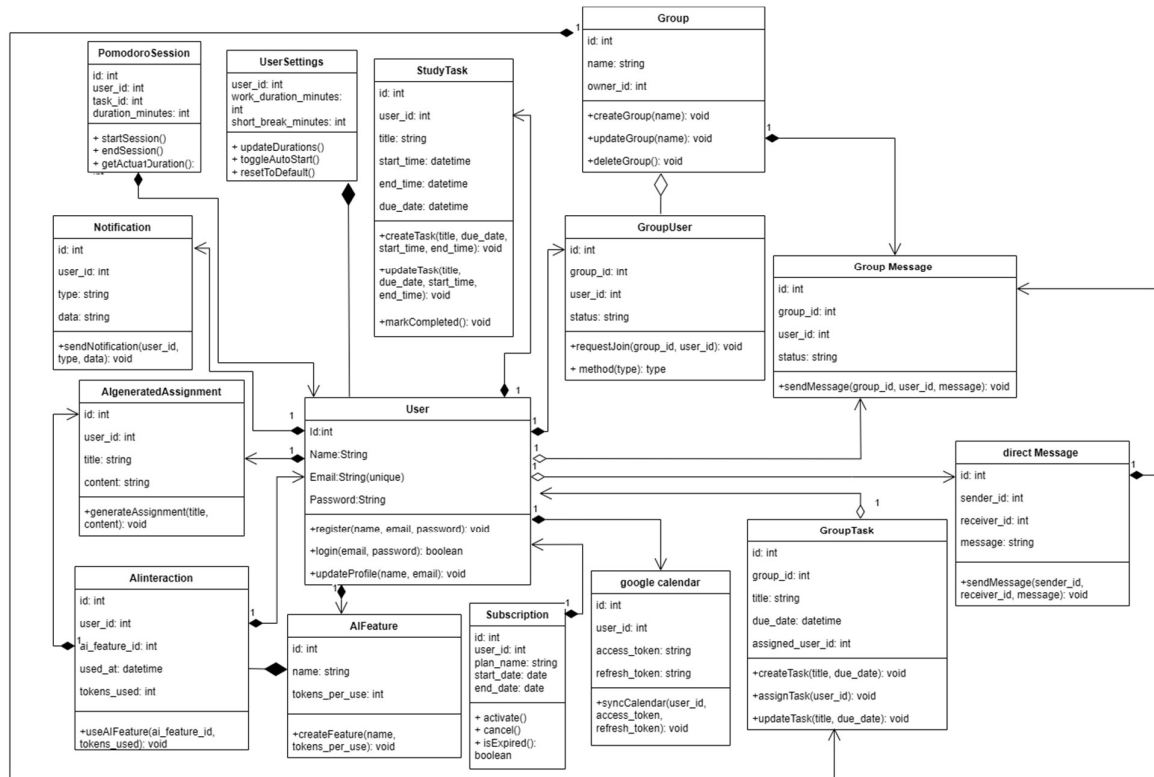


Figure 4.4 Class Diagram

4.3.5 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) visually represents the conceptual data model of the Uniflow system, illustrating the relationships between different key entities within the application's domain. It defines entities (such as User, Group, Task, Message, Assignment, Notification, Event), their attributes, and the cardinality of relationships (one-to-one, one-to-many, many-to-many), providing a foundational understanding of the data structure and integrity rules. This ERD forms the basis for the logical database schema.

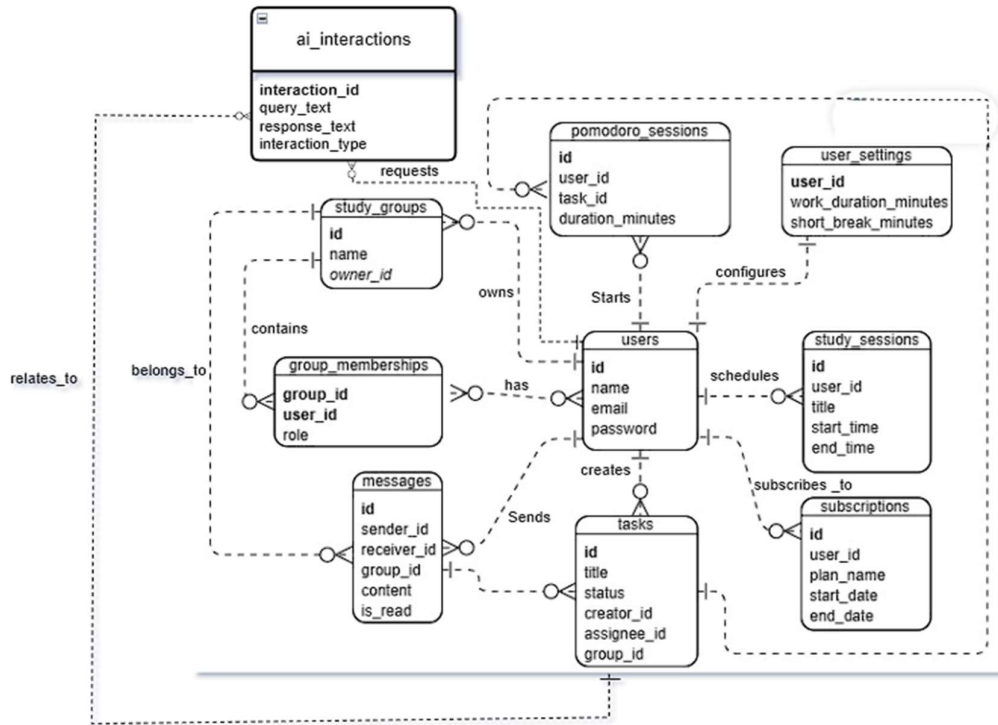


Figure 4.5 Entity Relationship diagram

4.4 System Design

The Uniflow system is designed as a modern web application following a service-oriented architecture pattern to ensure modularity, scalability, and maintainability. This section details the architectural choices, database structure, and user interface mock-ups, specifying how the system components will be constructed and interact.

4.4.1 System Architecture Diagram

This diagram provides a high-level visual representation of the Uniflow system's overall structure, illustrating the major components, their interactions, and the underlying technologies. It typically includes the frontend client, backend services (e.g., API server, AI processing server), database layer, caching layer, and integration points with external services such as the OpenRouter API and Google Calendar. A load balancer is depicted to manage incoming traffic.

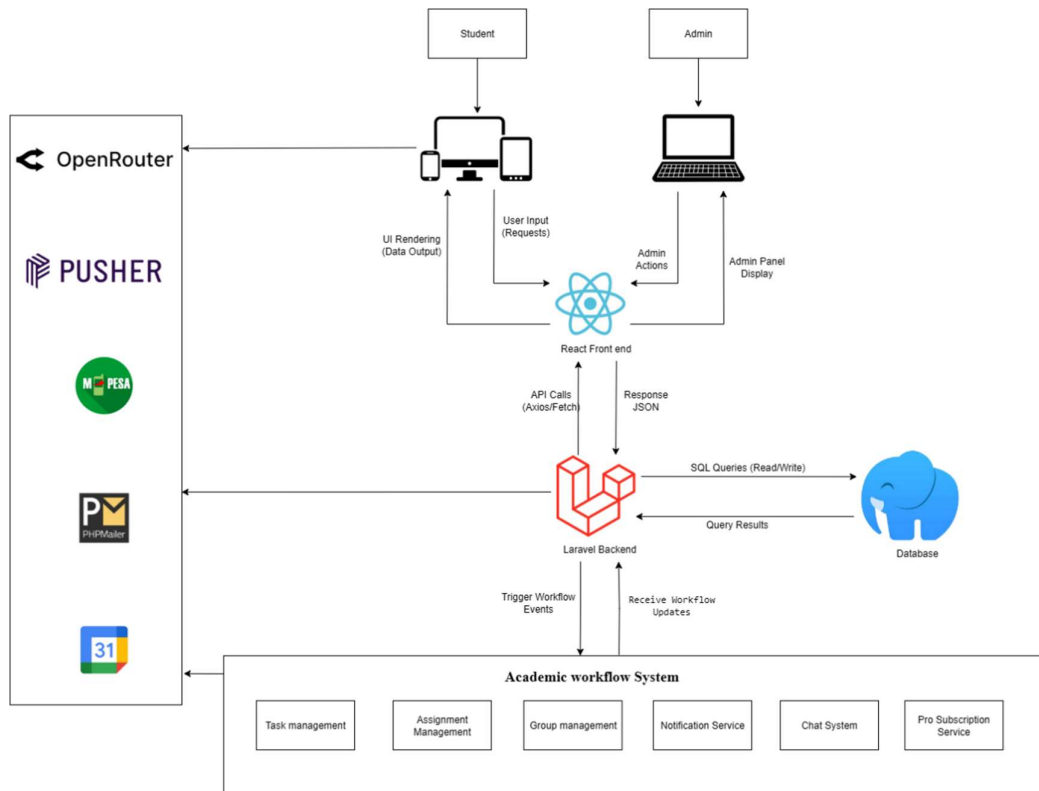


Figure 4.6 System Architecture

4.4.2 Database Schema Diagram

The Database Schema Diagram illustrates the logical data structure supporting the Uniflow application. Derived from the Entity Relationship Diagram (ERD) presented in the analysis phase, this schema shows the tables for users, groups, tasks, assignments, messages, notifications, and events. For each table, it details column names, data types, primary keys, foreign keys, and unique constraints, explicitly mapping out how data is organized for persistence and retrieval.

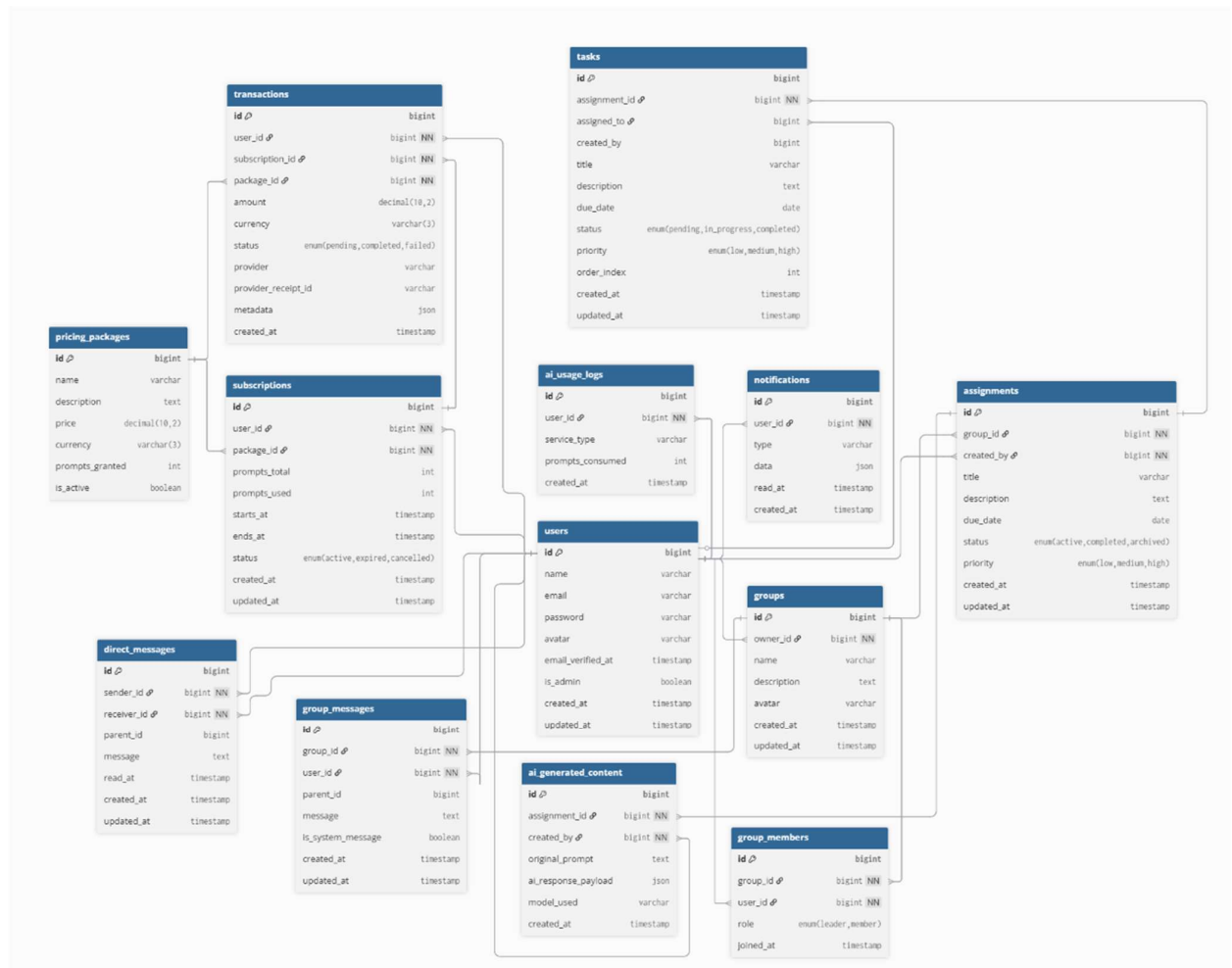


Figure 4.7 Database Schema

4.4.3 System Design Diagram

This diagram illustrates the overall architecture of the Uniflow system, showing the interaction between core components including the frontend interface, backend services, database, and third-party integrations such as WorkOS and Pusher. It defines data flow, module dependencies, and system boundaries, providing a high-level technical overview to guide implementation.

4.4.4 System Wireframes

These wireframes represent the skeletal user interface design for key Uniflow system screens. They focus on layout, information hierarchy, and user interaction flow, serving as visual blueprints for development. At least four distinct wireframes are provided.

4.4.4.1 Dashboard Wireframe

The dashboard wireframe provides a personalized overview for the student, displaying current academic tasks, upcoming deadlines, recent group activity, and quick access to frequently used features. It is structured with a left-hand navigation sidebar (e.g., My Tasks, Groups, Calendar, Messages, Settings), a main content area that can toggle between "Tasks Due Soon," "My Study Schedule," and "Recent Notifications," and customizable widgets for personalized academic insights (e.g., "Courses Progress," "Overdue Assignments"). Task cards within the display feature priority indicators, assigned collaborators, and clear deadlines, with click-through access to full task details. A prominent "Add New Task" button is positioned for easy access.

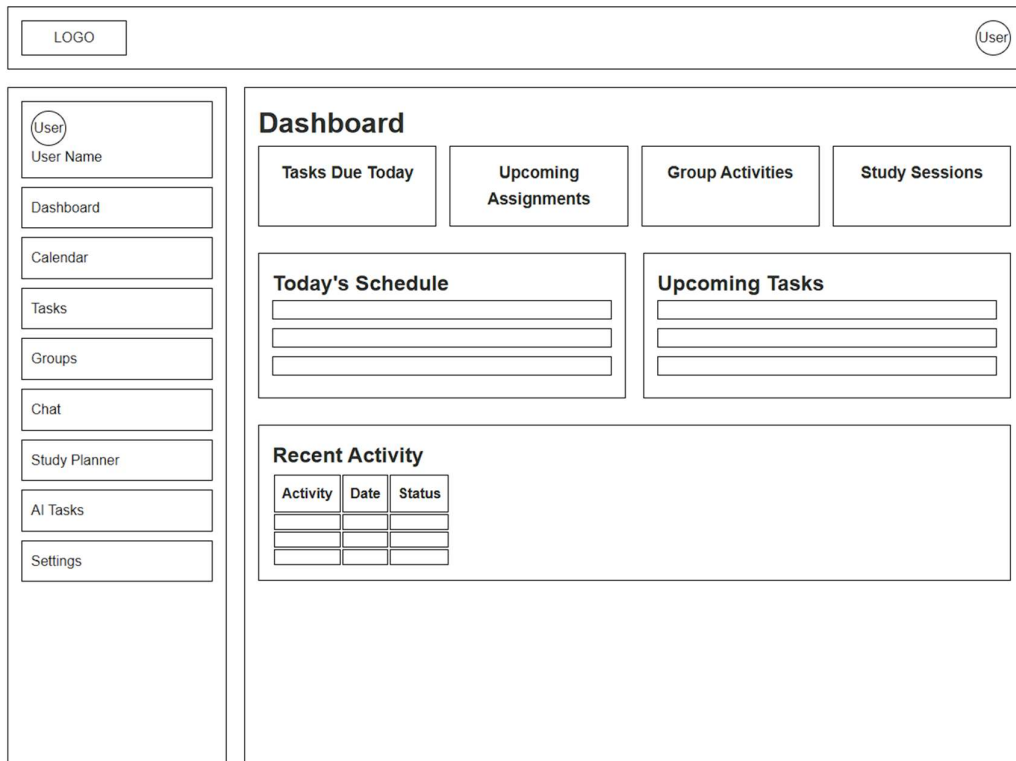


Figure 4.8 Dashboard wireframe

4.4.4.2 AI-Assisted Task Creation Wireframe

This wireframe illustrates the screen for AI-assisted task creation, allowing students to input natural language descriptions of their assignments. It features a large text input area (e.g., "Describe your assignment or project idea...") with instructional prompts. As the user types, a dynamic area labelled "AI Processing Status" (e.g., "Analysing

input...", "Generating tasks...") provides real-time feedback. Below the input, a "Generated Tasks Preview" section displays the AI's structured output as editable task items, complete with suggested titles, deadlines (with a date picker), proposed collaborators (with a dropdown for group members), and priority levels. Students can then modify any of these details, add or remove tasks, and reorder them before confirming with a "Review & Save" or "Discard" option.

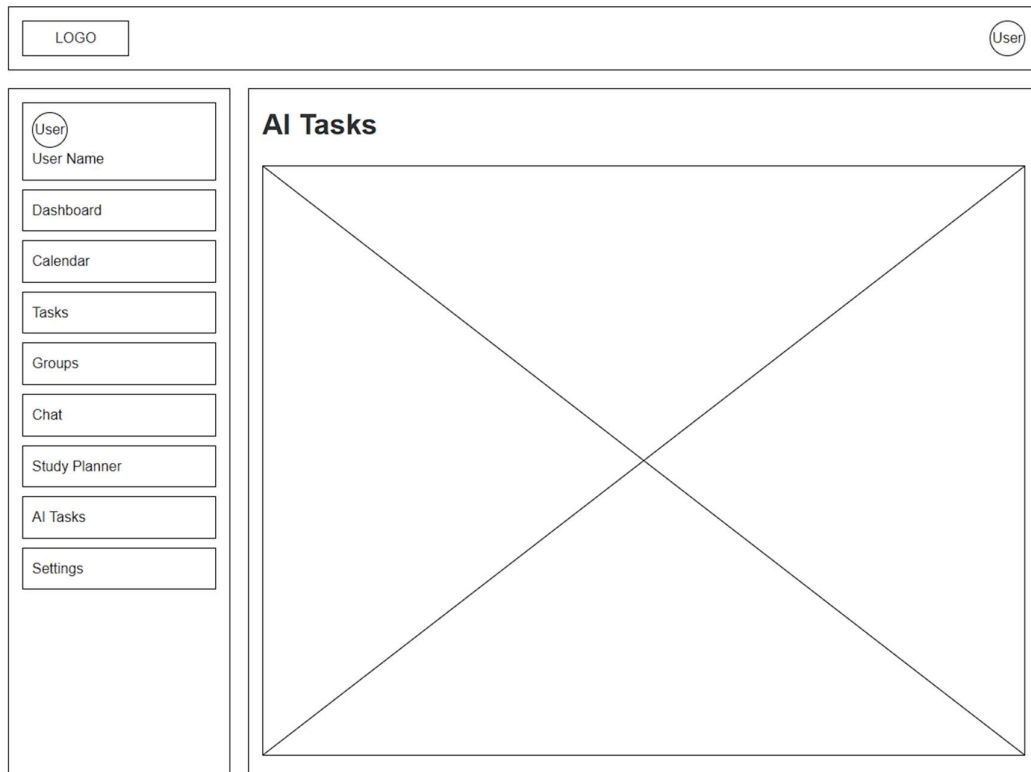


Figure 4.9 AI assisted task creation wireframe

4.4.4.3 Chat Interface Wireframe

The chat interface wireframe for academic collaboration presents a clean and intuitive layout for seamless real-time communication. The central area displays the ongoing chat history, featuring message bubbles with sender avatars, usernames, and timestamps. The input section at the bottom allows users to type messages, insert emojis, and attach files such as notes, assignments, or resources.

A right-hand sidebar shows active contacts or recent conversations, with presence indicators (online/offline) next to each user. The top section includes a search bar for quickly locating past messages, along with accessible tabs or icons for "Pinned

Messages" and "Shared Files," helping users manage key information efficiently. The layout supports both one-on-one and future group extensions while maintaining clarity and user-friendliness.

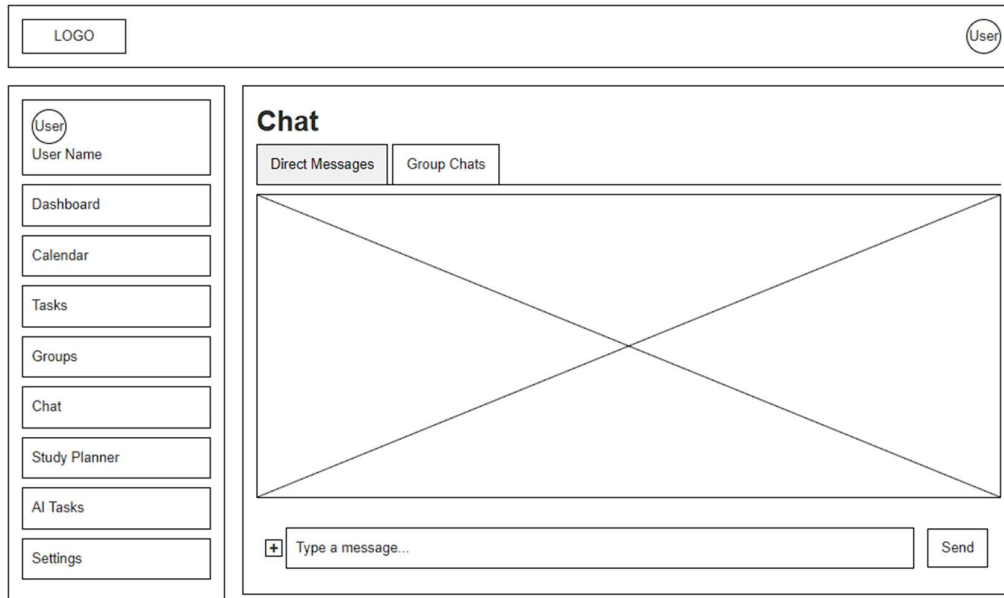


Figure 4.10 Chat Interface Wireframe

4.4.4.4 Calendar Wireframe

The calendar wireframe for the Study Planner module provides a streamlined interface for scheduling and managing academic activities. It features a customizable calendar view with options for daily, weekly, or monthly layouts. Students can easily block out time for study sessions, lectures, exams, and personal events.

A key feature is seamless integration with Google Calendar, supporting two-way synchronization. This ensures that any updates made within the study planner are reflected in the user's Google Calendar and vice versa. This integration allows students to manage their academic and personal schedules in one unified space, reducing the risk of conflicts and improving time management.

LOGO

User

User

User Name

Dashboard

Calendar

Tasks

Groups

Chat

Study Planner

AI Tasks

Settings

Calendar

Month

Week

Day

Agenda

Previous

June 2025

Next

Sync to Google Calendar

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21

Figure 4.11 Calendar Wireframe

Chapter 5: System Implementation and Testing

5.1 Introduction

This chapter details how the Web-Based Academic Workflow Management System was developed and examined through structured implementation and testing procedures. It transitions from the theoretical framework established in Chapter 4 to the practical application of development and evaluation principles. This section explains the technical setup used during development, including the chosen hardware and software tools. It also elaborates on the AI dataset employed and discusses the core implementation steps taken throughout the project. Lastly, it outlines a structured testing strategy aimed at verifying the system's reliability, efficiency, and security. The chapter concludes with a detailed analysis of the results gathered during the system validation phase.

5.2 Description of the Implementation Environment

This subsection aims to outline the hardware and software specifications used to build and operate the academic workflow system, ensuring adequate system efficiency and smooth operation. The information is organized into distinct sections to enhance readability and ensure clear documentation.

5.2.1 Hardware Specifications

The development and deployment of the system were facilitated by distinct hardware configurations tailored to the specific needs of each phase.

Table 5.1 Hardware Specification

Item	Testing Device Specification	Recommended Specification
RAM	16GB	8GB
Processor	Intel Core i7 (10th Gen)	Intel Core i5 (8th or 9th Gen)
Storage	512GB SSD (NVMe)	500GB SSD or HDD
System Type	64-bit Windows 11 Pro	64-bit Windows 10 or 11 (Home or Pro)

This configuration was sufficient for routine development tasks, including running local servers, compiling code, and managing multiple tools with minimal performance issues, provided it meets or exceeds the specifications outlined in the table.

5.2.2 Software Specifications

The software stack for this project was carefully chosen to utilize modern, efficient, and widely supported technologies for building a scalable and maintainable web-based academic workflow system. The frontend was developed using React.js (v18), selected for its component-based structure and high rendering performance, making it ideal for creating interactive user interfaces. Tailwind CSS was used to streamline interface styling through its utility-first approach, enabling rapid and consistent design. To manage complex application states such as real-time notifications and task updates, Redux Toolkit was integrated to provide a predictable and centralized state management solution. On the backend, Laravel 12 was adopted for its mature MVC architecture and built-in security capabilities. Authentication and frontend scaffolding were simplified through the Laravel WorkOS React Starter Kit, which includes pre-integrated support for WorkOS AUTH Kit and React components. The application runs on PHP 8.1, chosen for its performance improvements and modern features, hosted on an Nginx web server. MySQL 8.0 serves as the primary database for persistent storage needs. For real-time functionality, Laravel Echo was configured alongside Pusher to support live messaging and push notifications. The AI capabilities of the system are powered by OpenRouter's integration with the DeepSeek V3 instruction-tuned language model, which was selected for its effectiveness in understanding academic prompts and producing structured task outputs. Version control and collaborative development were managed using Git and GitHub, utilizing features such as pull requests and branching workflows. To maintain code quality and ensure consistent performance, GitHub Actions was implemented to perform continuous integration, automatically executing tests and code validations upon each commit. To support deployment and execution, the application should be hosted in a server or environment that meets or exceeds the recommended requirements for Laravel 12, including PHP 8.1 dependencies and compatibility with Laravel Echo.

Table 5.2 Server Requirements

Requirement	Description
PHP Version	PHP >= 8.1
Web Server	Nginx or Apache with mod_rewrite
Database	MySQL 8.0 or higher

Required Extensions	OpenSSL, PDO, Mbstring, Tokenizer, XML, JSON, Ctype
Node.js & npm	For building frontend assets
Composer	PHP dependency manager
Redis (optional)	For queueing and caching (optional)
Pusher (or equivalent)	For real-time broadcasting

5.3 Dataset Description

By interpreting high-level learning objectives, the system supports effective scheduling and prioritization of academic tasks, breaking them down into structured, actionable components. This capability is driven by the DeepSeek V3 instruction-optimized language model, which was fine-tuned using a curated dataset containing annotated academic frameworks, representative student submissions, and related educational artifacts. The system is capable of producing task structures when needed, arranging them into organized layers of overarching objectives and their corresponding subtasks to aid in smooth execution. After task creation, the system assigns them to group members using criteria such as complexity, team size, user preferences, and assigned skills to ensure balanced allocation. Notably, the task allocation remains adaptable, allowing the assignment logic to be reapplied even after tasks have been initially distributed. This enables both students and instructors to make real-time adjustments to the task plan depending on availability, timeline changes, or evolving project demands.

As part of the workload and complexity evaluation, each task includes two core indicators aimed at promoting clarity and equitable task sharing:

Workload Percentage shows how much of the overall project effort is assigned to each student, supporting balanced contributions and preventing excessive workloads.

Complexity Level indicates the relative difficulty of a task categorized as Low, Medium, or High to guide fair distribution and strategic task management.

These features collectively foster accountability, adaptive planning, and effective team collaboration key goals in academic project management.

5.4 System Implementation

The system is implemented as a modular web application using Laravel (backend), React (frontend), and MySQL (database). Core modules include secure authentication via WorkOS, real-time communication using Laravel Echo and Pusher, and a productivity suite with task management tools like Kanban boards and Gantt charts. The implementation follows Agile principles, focusing on real-time interactivity, security, and academic workflow optimization.

5.4.1 User Authentication and Profile Management

This foundational module establishes a secure and user-friendly gateway to the Uniflow ecosystem, powered by the enterprise-grade identity platform, WorkOS (AuthKit). Its implementation emphasizes security, flexibility, and personalization. Multi-provider SSO streamlines onboarding, allowing users to register and authenticate via traditional email/password or OAuth providers such as Google, Microsoft, Apple, and GitHub. This capability was thoroughly validated in test cases 4–13. Security is further enhanced with mandatory Two-Factor Authentication (2FA), requiring a time-based one-time password (TOTP) from authenticator apps like Microsoft Authenticator, as detailed in. Authenticated users have full control over their profiles, including the ability to update personal information and switch between light and dark appearance modes settings that persist across sessions. Secure session handling and password management are implemented using Laravel’s native security features, including CSRF protection and a token-based password reset workflow, as shown in Appendix 15.

5.4.2 Group Management and Real-Time Communication

This module forms the collaborative backbone of Uniflow by enabling structured teamwork and real-time communication. Users can create dedicated group workspaces for specific projects or courses, with creators automatically granted administrative privileges. Real-time messaging is facilitated through a chat engine built with Laravel Echo and Pusher, supporting both direct messaging and group-specific chat channels. Users experience fluid conversations without page reloads and can securely share files such as PDFs and images directly within chats, a feature validated in test case 20.

5.4.3 Academic Task and Productivity Suite

Designed as the core hub for student productivity, this module offers tools to effectively plan, visualize, and execute academic work. It supports a clear hierarchy of Assignments and their associated Tasks, allowing both macro- and micro-level project management. Tasks can be visualized in multiple formats, including Kanban boards for agile workflows, Gantt charts for time-sensitive projects, and simple to-do lists for straightforward task tracking. To promote focused study, a dedicated Study Planner is provided along with a built-in Pomodoro Timer, encouraging balanced work and rest sessions.

5.4.4 AI-Assisted Task Generation

A standout innovation within Uniflow, this module utilizes Artificial Intelligence to automate academic planning. Users input high-level assignment descriptions using natural language through a dedicated UI. These inputs are processed asynchronously via a queue job that sends the prompt to the DeepSeek V3 language model through the OpenRouter API, keeping the interface responsive. The AI then returns a structured JSON containing a detailed breakdown of tasks, suggested deadlines, priorities, and even recommended assignees. This was tested and validated with both simple and complex prompts in test cases 14 and 15.

5.4.5 Calendar and Scheduling Management

This module offers a centralized view of a student's academic life by aggregating tasks, deadlines, and study sessions into a unified calendar. A key feature is the two-way synchronization with Google Calendar, allowing events to be pushed both from Uniflow to Google and vice versa. This ensures that users have a single, consistent source of truth for their schedules across all devices, as verified in test case 18.

5.4.6 Subscription and Payment Integration

To enable access to premium features, such as advanced AI task generation, this module integrates a secure and localized payment system. Payments are facilitated via M-Pesa using the STK Push mechanism. A backend API endpoint handles M-Pesa callbacks, confirming successful transactions and upgrading user accounts automatically. The system handles various transaction states success, failure, and user-dismissal reliably, as demonstrated in test cases 22–27.

5.4.7 Administrative Dashboard and System Analytics

The administrative module provides role-based access to a centralized dashboard that functions as mission control for the platform. Administrators can view and manage key system entities, such as users, groups, and assignment. The dashboard includes advanced analytics, like registration trends, active users, task completion rates, AI usage and Mpesa payment analytics. It also monitors real-time system health, including database status and the availability of third-party services such as WorkOS and OpenRouter. Access is tightly controlled through role-based permissions, ensuring only authorized personnel can view and manage sensitive system-level data (validated in test case 2).

5.5 System Testing

An overview is provided of the robust testing methodologies adopted to validate the system's reliability, performance, and adherence to functional requirements. The testing process was structured to cover key areas such as functional accuracy, security flaws, performance efficiency, and overall user experience from various angles.

5.5.1 Testing Paradigm

The system testing followed a multi-layered approach, integrating various testing paradigms to ensure thorough evaluation across all key components:

White Box Testing was applied to examine internal code behaviour, monitor data paths, and assess code coverage across core modules such as user authentication, AI integration, and real-time messaging.

Black Box Testing: Applied to test user-facing functionality without knowledge of internal implementation, focusing on user workflows, API endpoints, and integration points.

Integration Testing: Conducted to verify seamless interaction between different system modules, particularly between the AI service, task management, and real-time communication components.

Performance Testing: Implemented to assess system responsiveness under various load conditions, especially for AI-powered task generation and real-time messaging features.

Security Testing: Performed to validate authentication mechanisms, authorization policies, and data protection measures across all user roles and access levels.

Table 5.3 Test Cases

Test Case	Description	Test Data	Expected Outcome	Actual Results	Status
1	Student registration with all fields	Full Name: Dhruvin Bhudia: dhruvin.bhudia@strathmore.edu	User registered, redirected to 2FA page	As expected	Pass As Shown in Appendix 4
2	Admin login with valid credentials	Email: admin@gmail.com Password: AdminPass123	Admin logged in , redirected to 2FA page	As expected	Pass As Shown in Appendix 5
3	2FA verification after registration	OTP successfully sent to the registered email address.	Valid OTP allows access to dashboard	As expected	Pass As Shown in Appendix 6
4	User login with valid credentials	Email: dhruvin.bhudia@strathmore.edu Password: denEnwOraNyt	Redirected to 2FA page before dashboard	As expected	Pass As Shown in Appendix 7
5	Invalid login attempt	Email: dhruvin@gmail.com Password: wrongpass	Error message shown	As expected	Pass As Shown in Appendix 8
6	Login with Google	Google login via OAuth	User authenticated	As expected	Pass As Shown in

			and redirected to dashboard		Appendix 9
7	Login with GitHub	GitHub account via OAuth	User authenticated and redirected to dashboard	As expected	Pass As Shown in Appendix 10
8	Login with Apple	Apple ID authentication	OAuth completes, access to dashboard	As expected	Pass As Shown in Appendix 11
9	Login with Microsoft	Microsoft OAuth login	OAuth completes, user redirected to dashboard	As expected	Pass As Shown in Appendix 12
10	2FA using Microsoft Authenticator	6-digit OTP from app	Code validated, user logged in	As expected	Pass As Shown in Appendix 13
11	2FA using Microsoft Authenticator	Invalid 6-digit OTP from app	Code rejected, user not yet logged in	As expected	Pass As Shown in Appendix 14
12	Password reset	Email: dhruvin@gmail.com	Reset link sent, password updated	As expected	Pass As Shown in Appendix 15
13	Group creation	Name: "web dev" Leader: Dhruvin	Group created, leader role assigned	As expected	Pass As Shown in Appendix 16

14	AI task generation (basic)	Prompt: “Create 3 web dev tasks”	Structured task data returned	As expected	Pass As Shown in Appendix 17
15	AI task generation (complex)	Prompt: “Create ML project with review, data collection”	Multi-layer task structure returned	As expected	Pass As Shown in Appendix 18
16	Task creation and assignment	Task title, description, deadline, assignee	Task saved and assigned	As expected	Pass As Shown in Appendix 19
17	Calendar task display	Task with due date and time	Displayed on correct calendar slot	As expected	Pass As Shown in Appendix 20
18	Google Calendar sync	User connects Google account	Tasks synced with user's calendar view	As expected	Pass As Shown in Appendix 21
19	Direct messaging	Message between two users	Real-time delivery and storage	As expected	Pass As Shown in Appendix 22
20	File sharing in chat	Files: PDF, JPG <10MB	File uploaded, link visible in chat	As expected	Pass As Shown in Appendix 23
21	Accessibility test	Keyboard navigation + screen reader	Labels accessible,	Minor issues	Partial Pass

			minor improvement needed		
22	M-Pesa payment	Valid Phone: 254112916170	M-Pesa prompt (STK Push) is triggered	As expected	Pass As Shown in Appendix 24
23	M-Pesa payment	Invalid Phone: 0712ABCD / too short	No prompt sent; validation fails	As expected	Pass As Shown in Appendix 25
24	M-Pesa payment	Student dismisses prompt on phone	Payment fails, system receives failure status	As expected	Pass As Shown in Appendix 26
25	M-Pesa payment	Enter correct PIN in prompt	Payment accepted, prompt count credited	As expected	Pass As Shown in Appendix 27

5.5.3 Testing Coverage Summary

The conducted test cases effectively validate the core functionalities of the application across various user roles and essential workflows. The registration process for the students was successfully tested, with proper redirection to the two-factor authentication (2FA) page using Microsoft Authenticator. Both valid and invalid OTP scenarios were handled as expected, ensuring that only verified users gain access to the dashboard. Login functionality was confirmed to support standard email/password authentication alongside third-party sign-ins through OAuth providers such as Google, GitHub, Apple, and Microsoft. In each case, users were redirected through the appropriate authentication flow, followed by 2FA verification before gaining dashboard

access. Each authentication method correctly redirected users through its respective flow, with successful enforcement of 2FA before access to the dashboard was granted.

The password recovery process was tested to ensure reset emails were dispatched and new credentials were applied accurately upon submission. The AI task generation functionality was tested with both basic and complex prompts, producing structured and context-aware task lists. Group creation, manual task assignment, and distribution processes were also verified, ensuring collaborative group features operate seamlessly. Real-time communication features including one-on-one messaging and document sharing were verified to operate smoothly, with consistent delivery and proper archiving of both messages and files. The Google Calendar integration was successfully tested, confirming that tasks sync accurately to users' calendar views. Calendar-based visualizations of tasks displayed entries on the correct dates.

Accessibility testing revealed basic compliance with screen reader and keyboard navigation requirements, though some minor improvements are still needed for full WCAG compliance. Out of the 25 test cases, 24 passed completely, with 1 marked as "Partial Pass" due to minor accessibility issues. This demonstrates that the application meets high standards for functional completeness, security, and usability, with strong integration across third-party services and essential user workflows.

Chapter 6: Conclusions, Recommendations and Future Works

6.1 Conclusions

Uniflow achieved its core goal of delivering a unified academic workflow solution designed with student needs at its centre. The system integrates task management, group collaboration, and AI-assisted planning into one intuitive interface. Users can create tasks manually or via AI, collaborate in real time, sync calendars, and manage workloads all contributing to a less stressful academic experience. Despite challenges like limited time, reliance on third-party APIs, and English-only support, the project demonstrated that even a small, focused team can deliver meaningful innovation. The platform notably enhanced both solo task management and team collaboration, aligning with a user-first philosophy in academic planning.

6.2 Recommendations

Institutions adopting Uniflow should host it on scalable cloud platforms like AWS or Azure with strong security protocols including 2FA, regular penetration testing, and encrypted communication. Effective onboarding (via tutorials and workshops) and showcasing practical AI use cases will drive engagement. Monitoring third-party services like OpenRouter and implementing fallback strategies is critical for uptime. Behaviour-based UX enhancements should be prioritized, supported by analytics, and the system should be containerized (e.g., Docker) to simplify mobile expansion. Accessibility should be embedded into development cycles rather than added post-hoc.

6.3 Future Works

Upcoming Uniflow enhancements will prioritize mobile applications with offline functionality, expanded AI support through model hubs and custom fine-tuning, and tighter LMS connectivity using frameworks like IMS Global. Additional upgrades will feature mental load detection and focused study environments to promote healthier, more productive learning habits. To encourage wider adoption, Uniflow will introduce multilingual interfaces, region-specific academic calendars, and culturally adaptive planning tools. An intelligent academic dashboard is also under development, designed to provide predictive trends and highlight areas where students may need additional support. Long-term visions include campus API integrations and blockchain credentialing, establishing Uniflow as the digital backbone of academic life.

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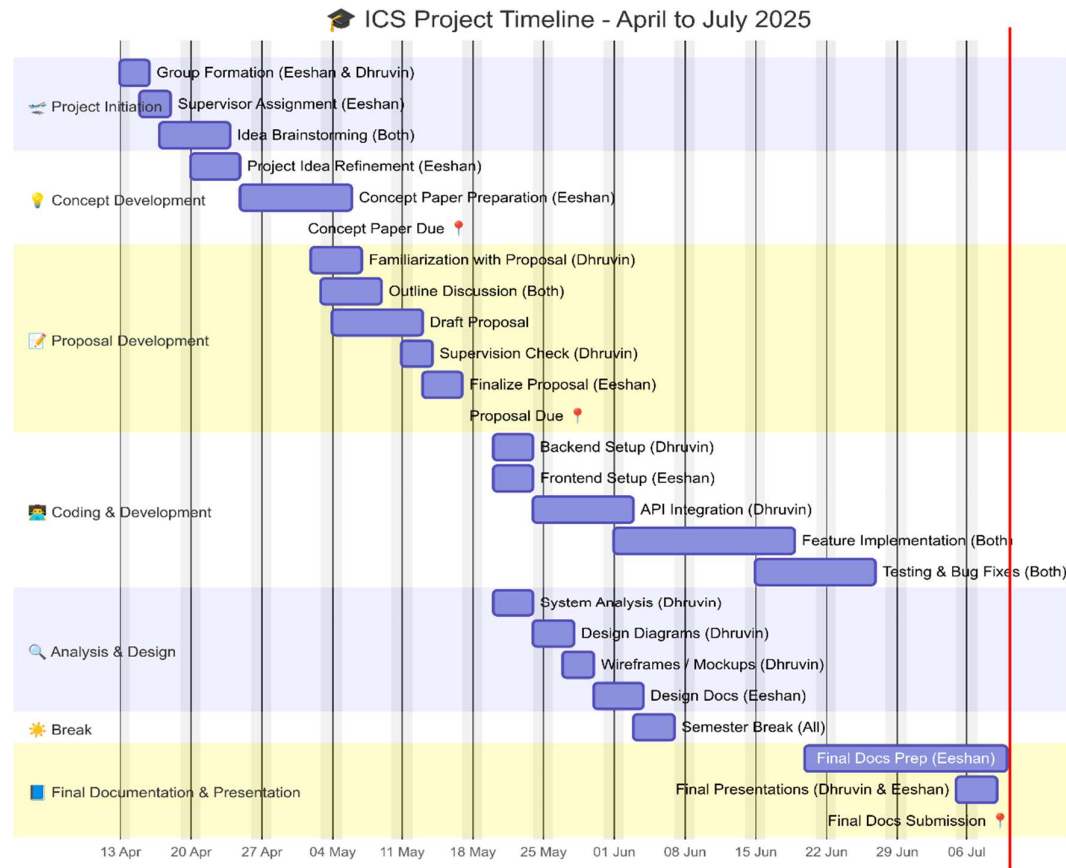
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<https://doi.org/10.1080/09687769.2024.2178654>
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<https://www.zendesk.com/blog/workload-analysis>

Appendix

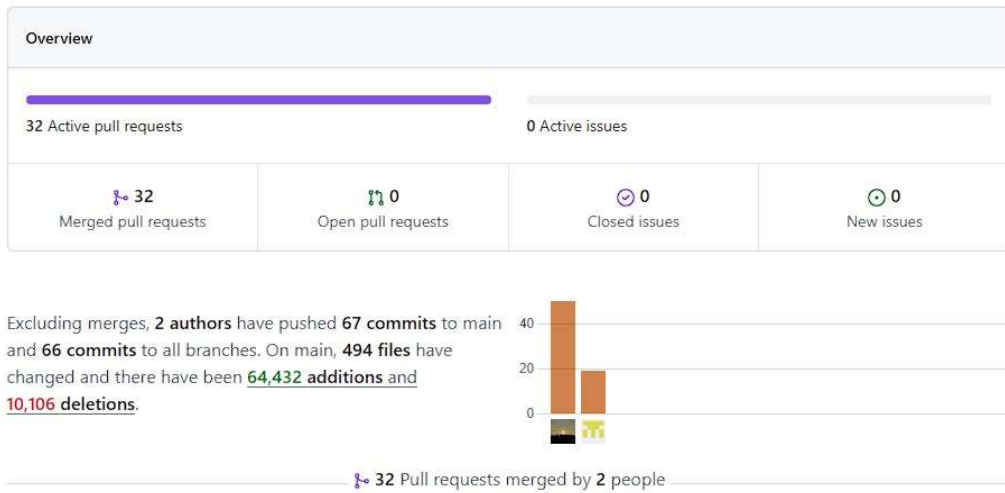
Appendix 1 : Gantt Chart



Appendix 2 Github Pull Insights

June 1, 2025 – July 1, 2025

Period: 1 month ▾



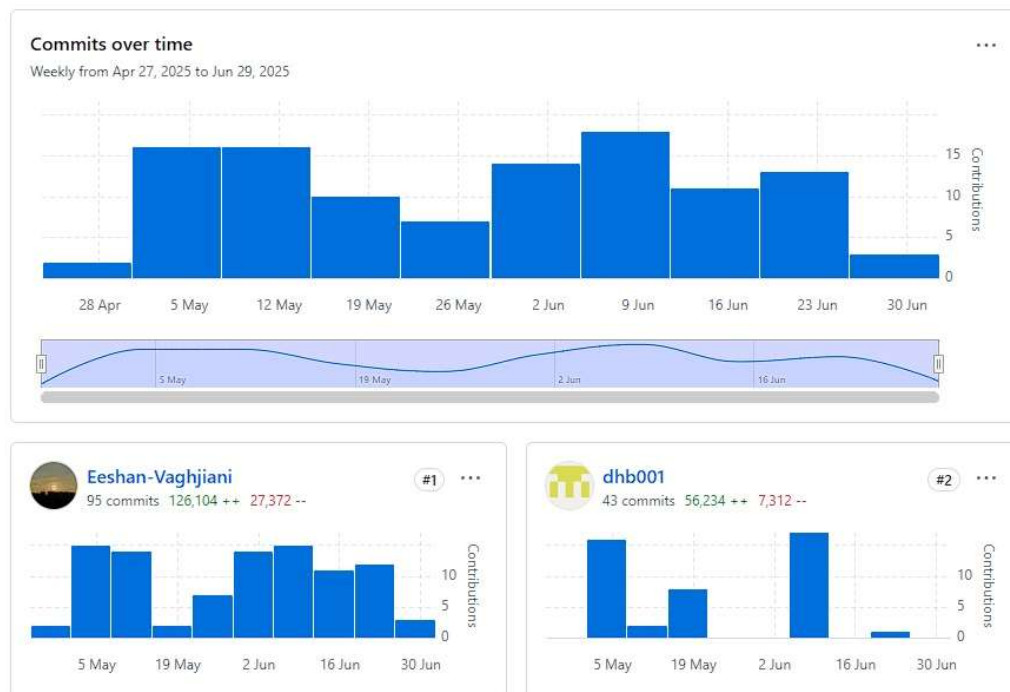
Appendix 3 GitHub Commits Graph

Contributors

Period: All ▾

Contributions: Commits ▾

Contributions per week to main, excluding merge commits



Appendix 4 New User Registration

Verify your email

Enter the code sent to
dhruvin.bhudia@strathmore.edu

[← Back to sign-in](#)

Appendix 5 Admin login with valid credentials

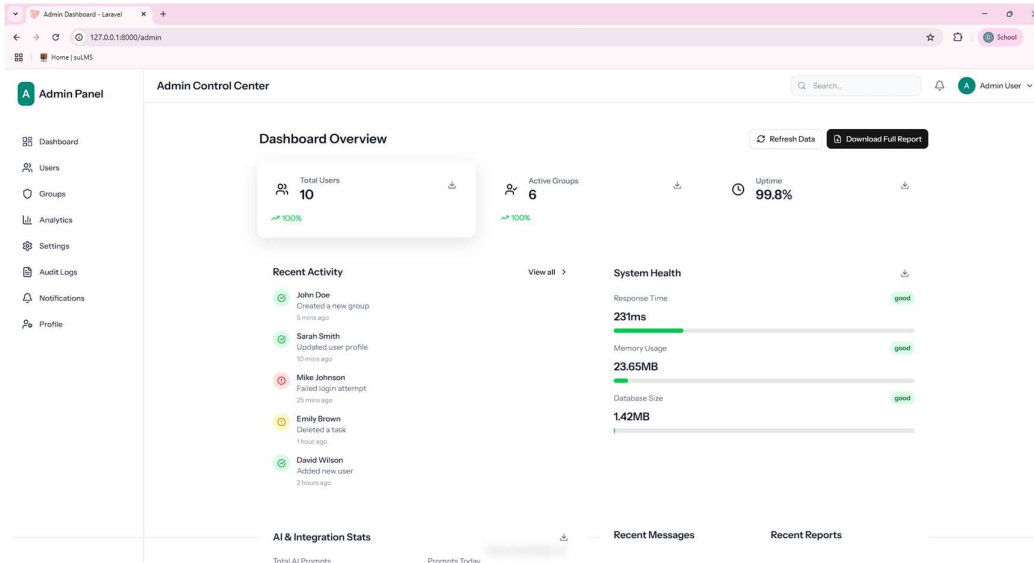
Sign in

Email
admin@gmail.com

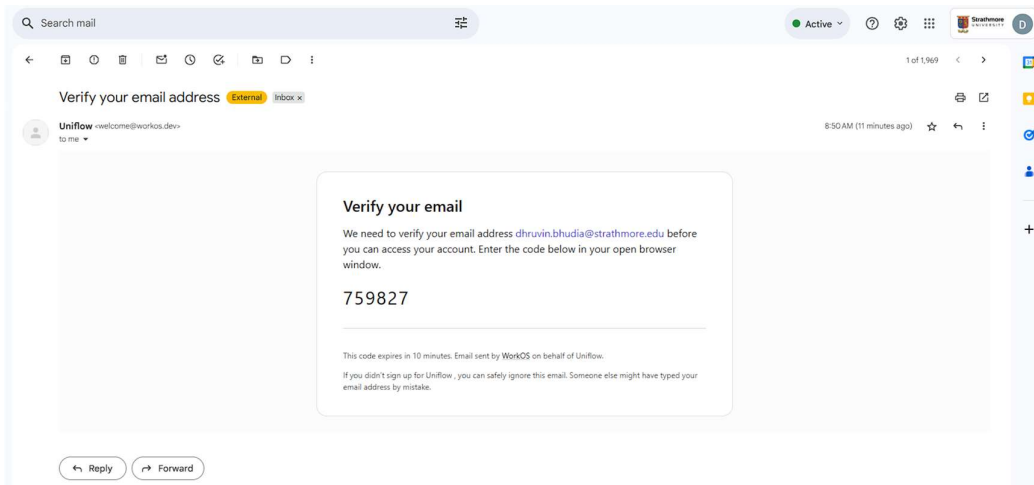
Password [Forgot your password?](#)
.....

OR

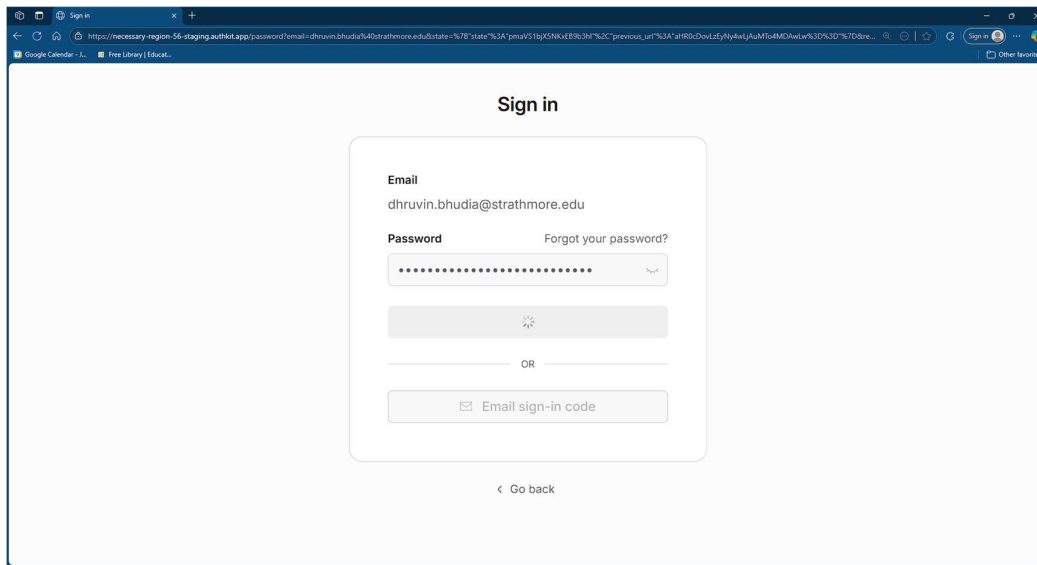
[← Go back](#)



Appendix 6 2FA verification after registration



Appendix 7 User login with valid credentials



The screenshot shows a web browser window with the URL `https://necessary-region-56-staging-authkit.app/password?email=dhruvin.bhudia%40strathmore.edu&state=%7B%22state%3A%22%3B%22previous_url%3A%22%3B%22%7D&next=...`. The page is titled "Sign in" and features a central login form. The form has an "Email" field with the value `dhruvin.bhudia@strathmore.edu` and a "Password" field with masked characters. A "Forgot your password?" link is next to the password field. Below the password field is a button with a loading spinner. An "OR" separator is followed by an "Email sign-in code" button. At the bottom of the form is a "Go back" link.

Sign in

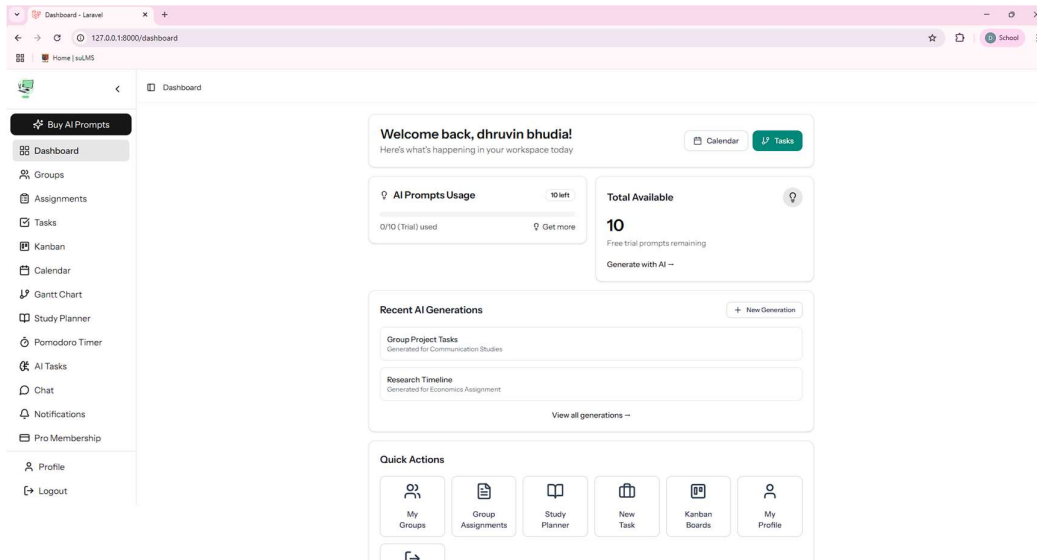
Email
dhruvin.bhudia@strathmore.edu

Password [Forgot your password?](#)
.....

OR

Email sign-in code

[Go back](#)



The screenshot shows a web browser window with the URL `127.0.0.1:8000/dashboard`. The page is titled "Dashboard" and features a sidebar with navigation links: Dashboard, Groups, Assignments, Tasks, Kanban, Calendar, Gantt Chart, Study Planner, Pomodoro Timer, AI Tasks, Chat, Notifications, Pro Membership, Profile, and Logout. The main content area displays a "Welcome back, dhruvin bhudia!" message. Below the welcome message are two cards: "AI Prompts Usage" showing 10 left and 0/10 (trial) used, and "Total Available" showing 10 free trial prompts remaining. Below these cards is a "Recent AI Generations" section with two entries: "Group Project Tasks" and "Research Timeline". At the bottom is a "Quick Actions" section with six buttons: My Groups, Group Assignments, Study Planner, New Task, Kanban Boards, and My Profile.

Dashboard - Laravel

127.0.0.1:8000/dashboard

Home | xLMS

Dashboard

Buy AI Prompts

Dashboard

Groups

Assignments

Tasks

Kanban

Calendar

Gantt Chart

Study Planner

Pomodoro Timer

AI Tasks

Chat

Notifications

Pro Membership

Profile

Logout

Welcome back, dhruvin bhudia!

Here's what's happening in your workspace today

Calendar Tasks

AI Prompts Usage 10 left

0/10 (trial) used Get more

Total Available

10

Free trial prompts remaining

Generate with AI --

Recent AI Generations

Group Project Tasks

Generated for Communication Studies

Research Timeline

Generated for Economics Assignment

View all generations --

Quick Actions

My Groups

Group Assignments

Study Planner

New Task

Kanban Boards

My Profile

[>]

Appendix 8 Invalid login attempt

Sign in

ⓘ Invalid email or password

Email
dhruvin@gmail.com

Password [Forgot your password?](#)
wrongpass

Sign in

OR

✉ Email sign-in code

[Go back](#)

Appendix 9 Login with Google

Sign in - Google accounts

accounts.google.com/o/oauth2/v2/auth/oauthchooseaccount?access_type=offline&scope=https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email%20https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.profile&state=eyJ... Verily it's you

Home | suLMS

Sign in with Google

Choose an account

to continue to [workos.com](#)

 Dhruvin Bhudia
dhruvin.bhudia@strathmore.edu

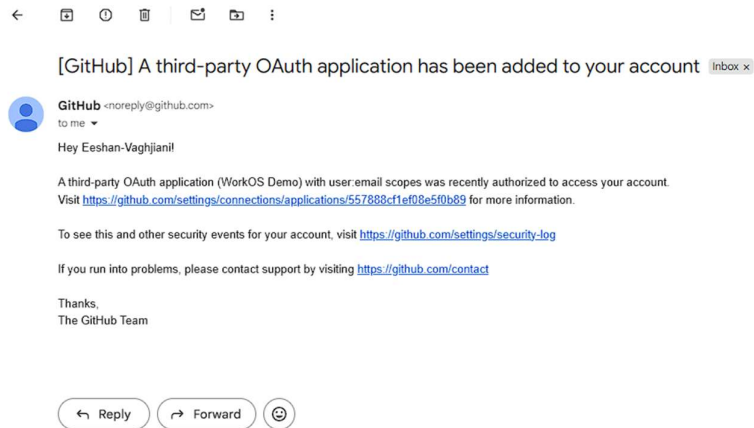
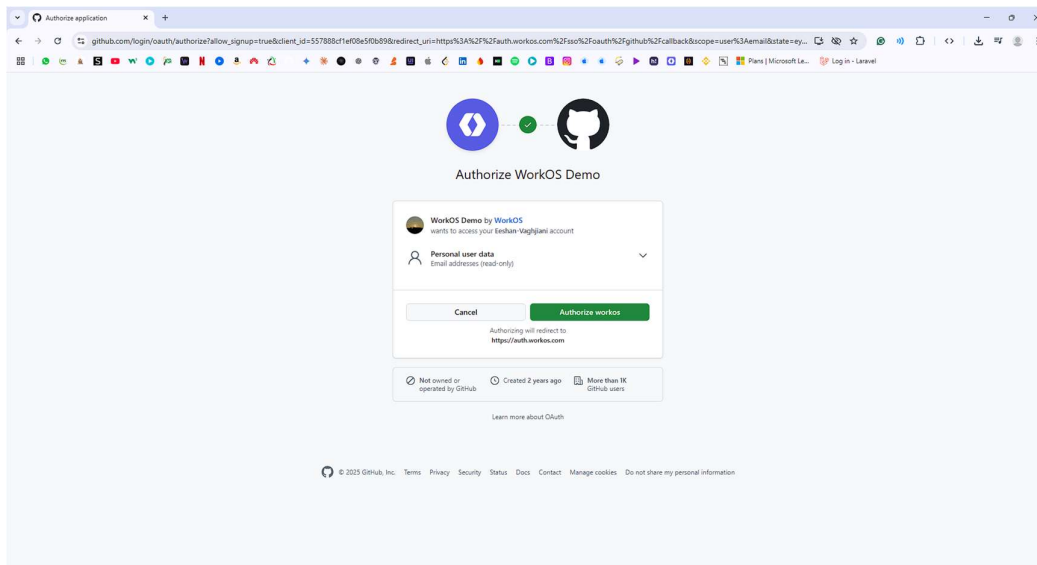
 Use another account

Before using this app, you can review [workos.com's privacy policy](#) and [Terms of Service](#).

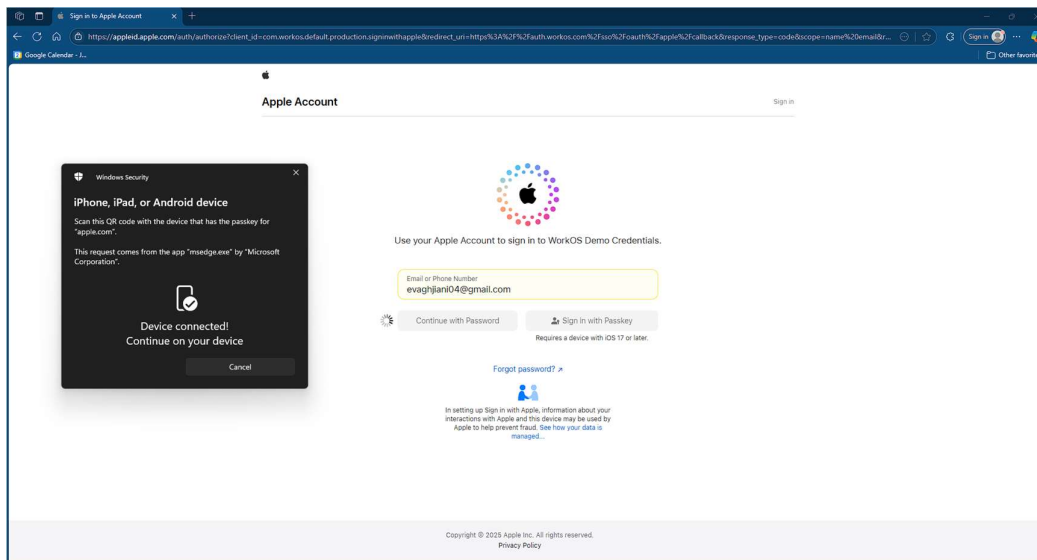
English (United Kingdom)

[Help](#) [Privacy](#) [Terms](#)

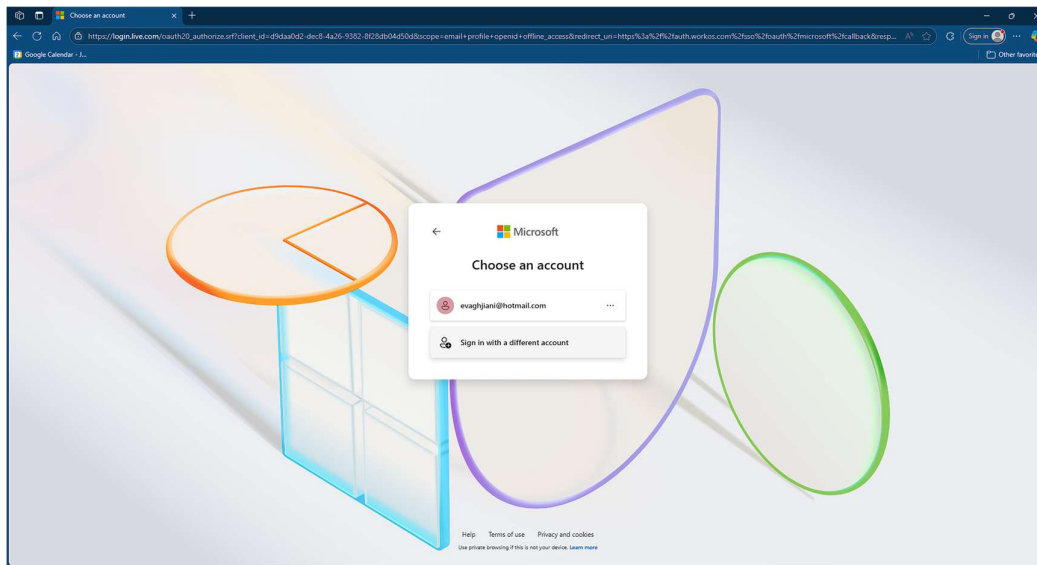
Appendix 10 Login with GitHub



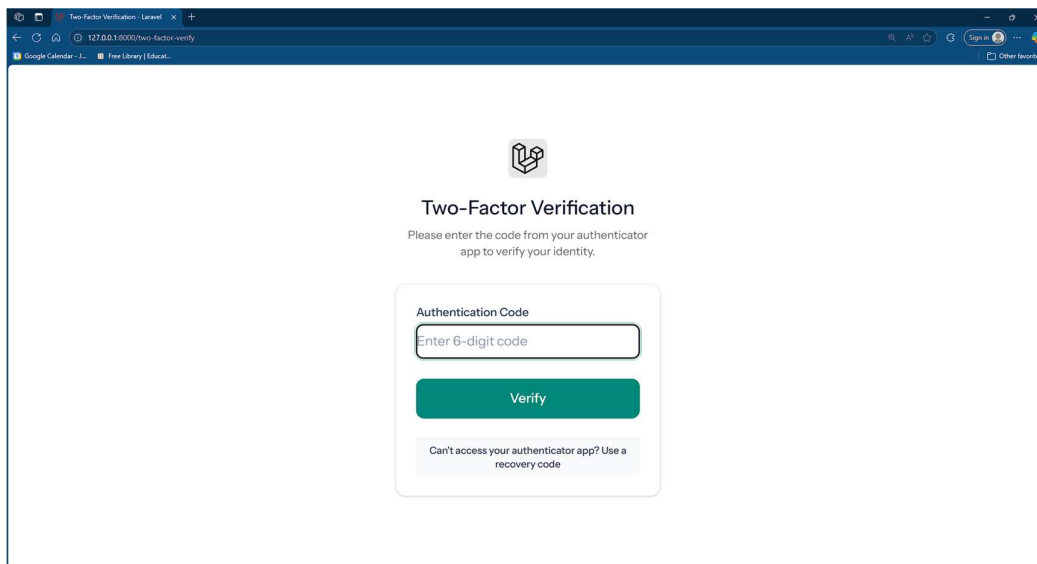
Appendix 11 Login with Apple



Appendix 12 Login with Microsoft

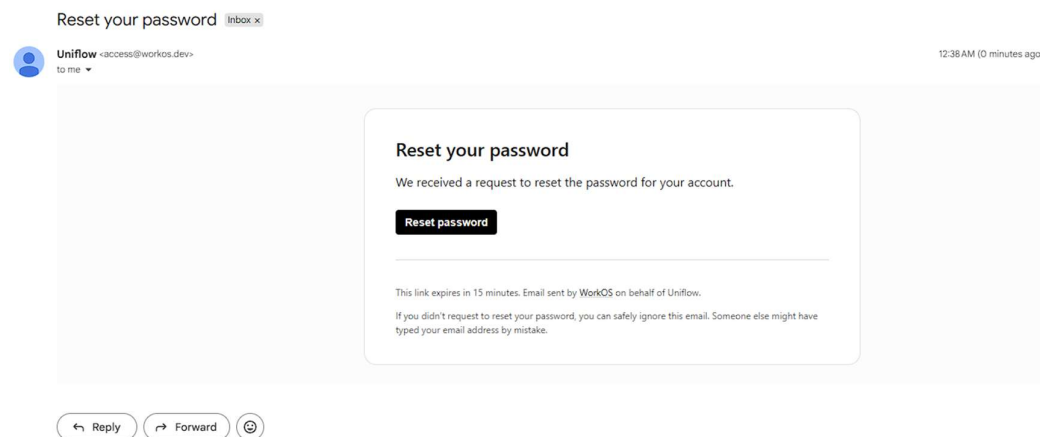


Appendix 13 2FA using Microsoft Authenticator

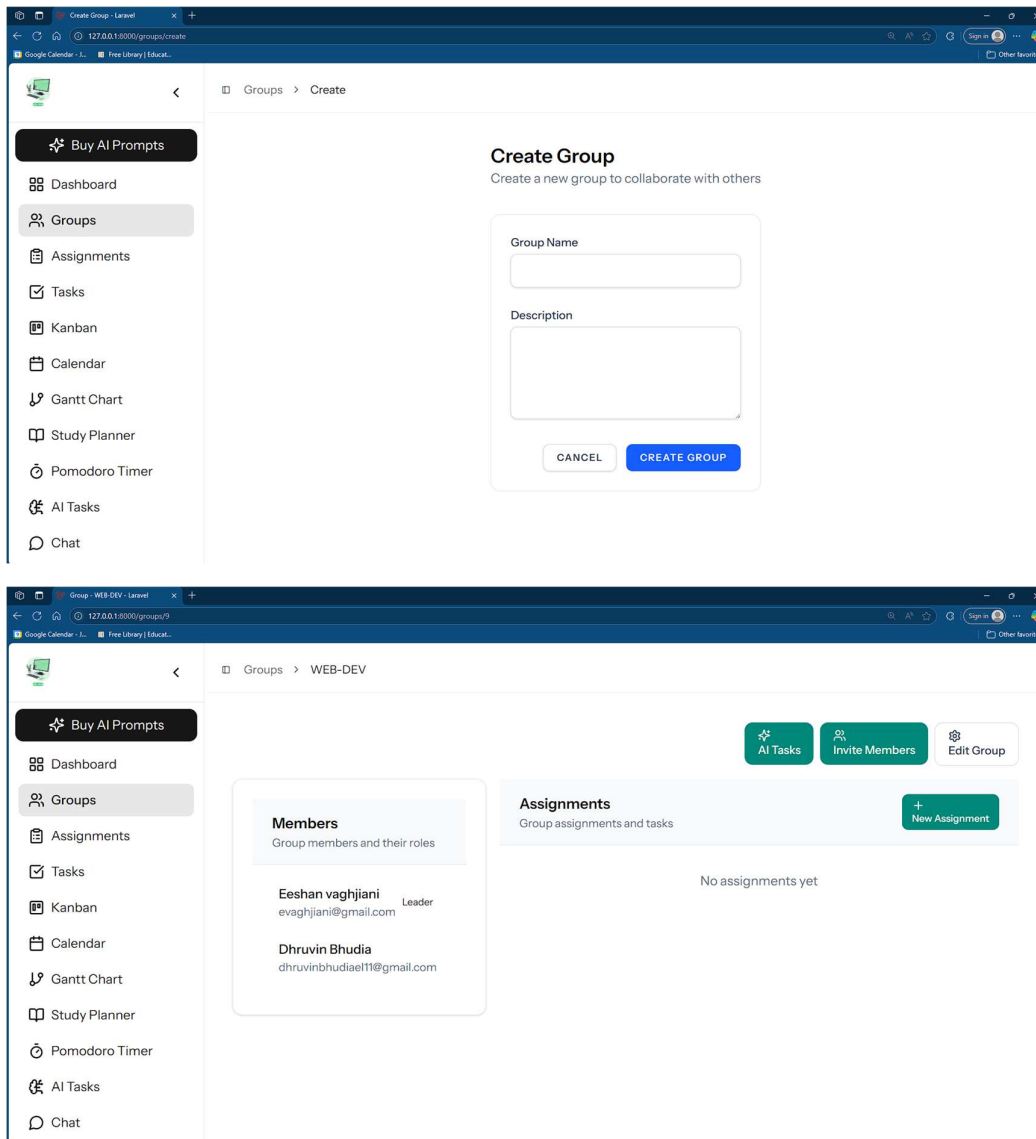


Appendix 14 invalid 2FA using Microsoft Authenticator

Appendix 15 Password reset

[illegible]

Appendix 16 Group creation



Appendix 17 AI task generation (basic)

Buy AI Prompts

Dashboard

Groups

Assignments

Tasks

Kanban

Calendar

Gantt Chart

Study Planner

Pomodoro Timer

AI Tasks

Chat

Notifications

Pro Membership

Profile

Logout

Groups > web.dev > AI Task Assignment

Let AI help you create and distribute tasks

Create New Assignment with AI

Describe the assignment and tasks in detail. The AI will generate structured tasks based on your description and distribute them among team members.

Create 3 web dev tasks

Be specific about what the assignment entails, including requirements, format, learning objectives, and any other relevant details.

Press **Ctrl+Enter** to submit

Web Development Assignment

C3351 - Advanced Web Development

Description

Develop a dynamic web application featuring a user-friendly interface, robust backend, and seamless user experience.

Due: 15/07/2025

Generated Tasks

Redistribute Tasks

Task	Assigned To	Due Date	Priority	Effort	Importance
Design Wireframes Create detailed wireframes for the web application's layout and user interface.	dhruv.in.bhudia	06/07/2025	high	4h	5/5
Develop Backend APIs Design and implement backend APIs for interacting with the web application's data and functions.	dhruv.in.bhudia	10/07/2025	medium	7h	4/5
Implement Frontend Components Code the frontend components, ensuring they are responsive and user-friendly.	dhruv.in.bhudia	15/07/2025	low	6h	3/5

Appendix 18 AI task generation (complex)

Groups > web.dev > AI Task Assignment

Buy AI Prompts

- Dashboard
- Groups
- Assignments
- Tasks
- Kanban
- Calendar
- Gantt Chart
- Study Planner
- Pomodoro Timer
- AI Tasks
- Chat
- Notifications
- Pro Membership
- Profile
- Logout

Let AI help you create and distribute tasks

Create New Assignment with AI
Describe the assignment and tasks in detail. The AI will generate structured tasks based on your description and distribute them among team members.
Create ML project with review, data collection

Be specific about what the assignment entails, including requirements, format, learning objectives, and any other relevant details. Press **Ctrl+Enter** to submit

ML Project with Review and Data Collection
Data Science
Description
Develop a machine learning project, including data collection and review stages.
Due: 01/10/2025

Generated Tasks [Redistribute Tasks](#)

Task	Assigned To	Due Date	Priority	Effort	Importance
Data Collection Strategy Create a strategy for data collection required for the ML project.	dheerav.bhudia	01/01/2025	High	8h 5/5	
Sample Data Collection Collect data samples required for the ML project.	dheerav.bhudia	14/01/2025	Medium	6h 4/5	
Data Cleaning and Preprocessing Clean and preprocess the collected data for ML model.	dheerav.bhudia	20/01/2025	Medium	4h 3/5	
ML Model Design Design a machine learning model appropriate for the data.	dheerav.bhudia	23/01/2025	High	5h 5/5	
Train and Test Model Train and test the designed machine learning model.	dheerav.bhudia	03/06/2025	High	5h 4/5	
Model Improvement and Hyperparameter Tuning Improve and tune the machine learning model as necessary.	dheerav.bhudia	10/06/2025	Medium	4h 3/5	
ML Project Presentation Prepare and deliver a presentation on the ML project and its results.	dheerav.bhudia	17/06/2025	High	3h 5/5	

Appendix 19 Task creation and assignment

IS Project Final Documentation: x +

127.0.0.1:8000/groups/7/assignments/22

Home | suLMS

Assignments > IS Project Final Documentation

Buy AI Prompts

- Dashboard
- Groups
- Assignments
- Tasks
- Kanban
- Calendar
- Gantt Chart
- Study Planner
- Pomodoro Timer
- AI Tasks
- Chat
- Notifications
- Pro Membership
- Profile
- Logout

IS Project Final Documentation [EDIT](#) [EDIT AI ASSIGNMENT](#) [DELETE](#) [ADD TASK](#) [TASK ASSIGNMENTS](#)

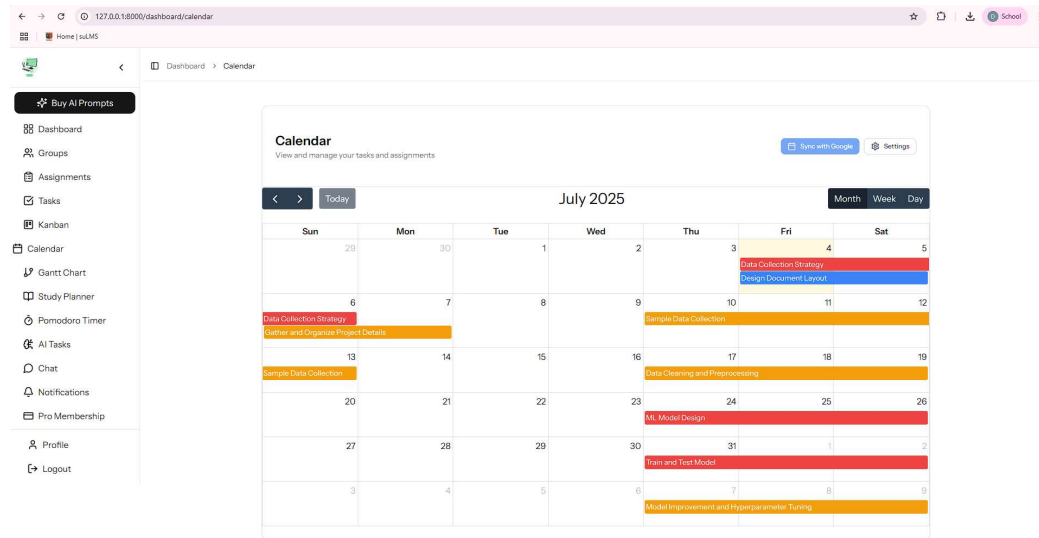
web.dev - Information Systems

Preparing the final documentation for the IS project
Due: 11/30/2025 **Medium Priority**

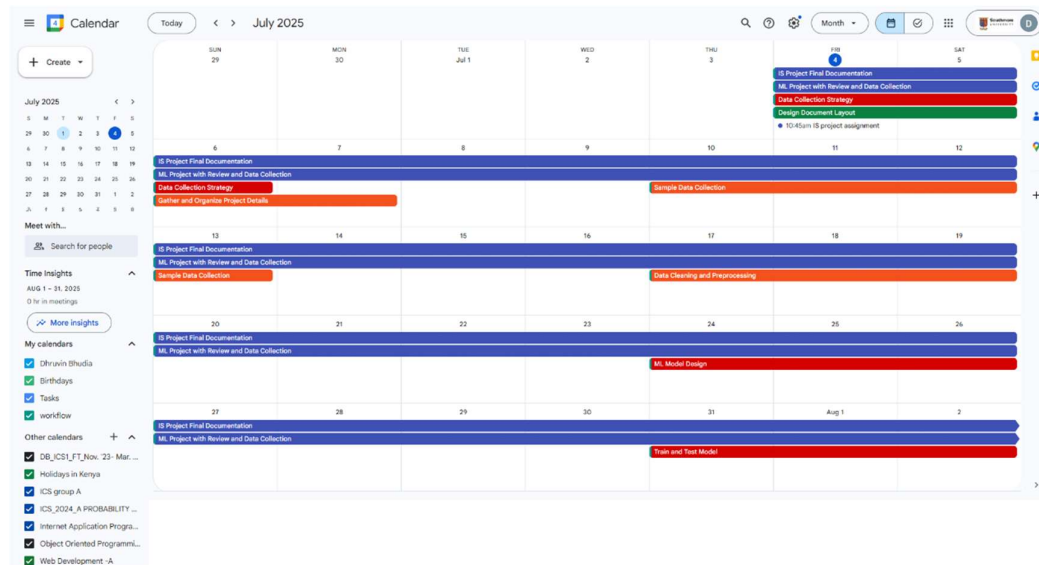
Tasks

Design Document Layout Develop a visually appealing and easy-to-understand layout for the project documentation Due: 7/6/2025 Low Priority Pending	COMPLETE
Gather and Organize Project Details Collect and categorize all necessary project details for inclusion in the documentation Due: 7/6/2025 Medium Priority Pending	COMPLETE

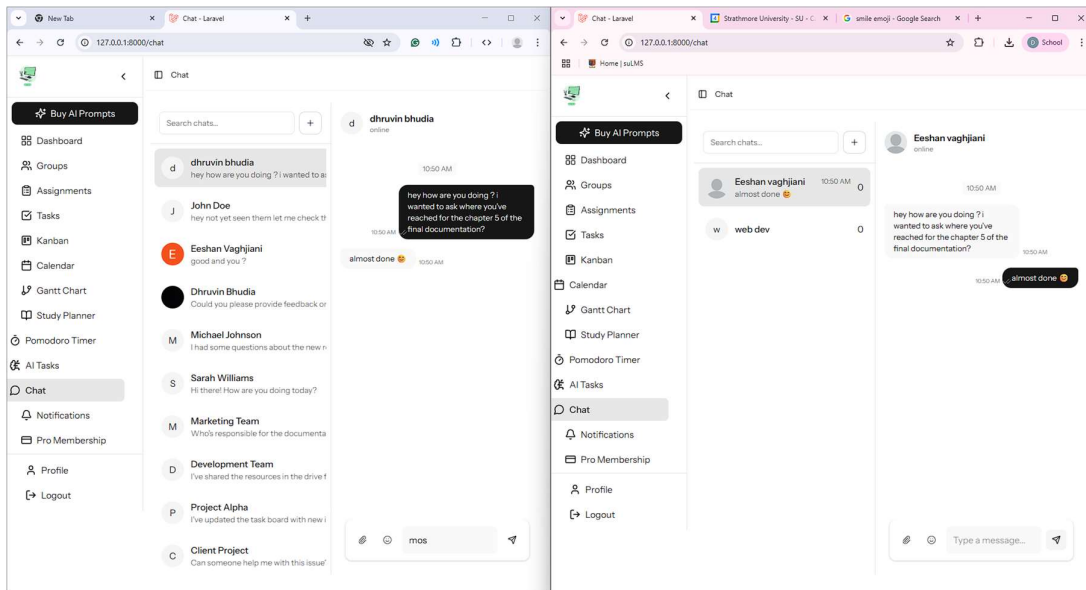
Appendix 20 Calendar task display



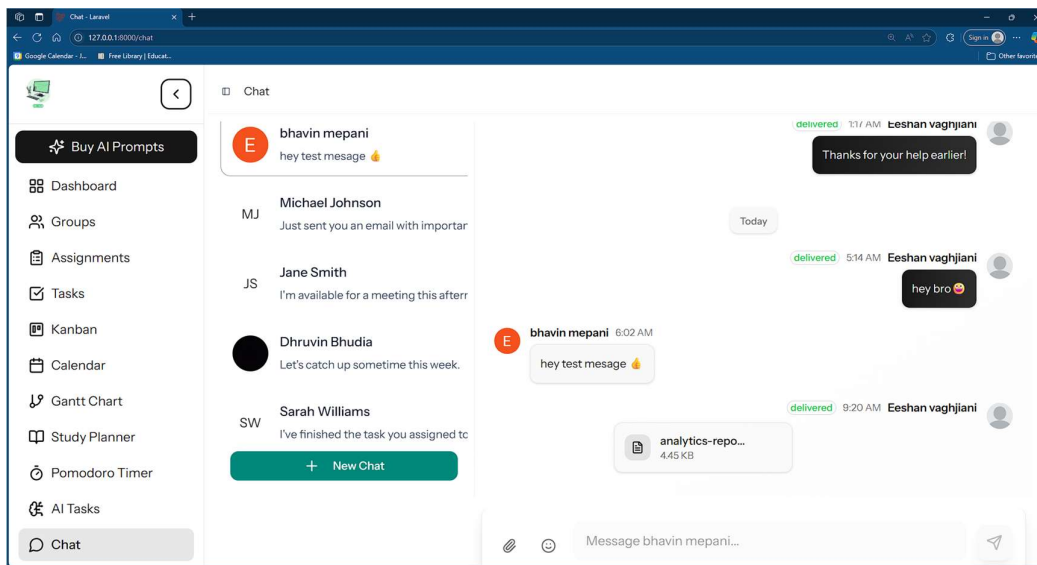
Appendix 21 Google Calendar sync



Appendix 22 Real time Direct messaging



Appendix 23 File sharing in chat



Appendix 24 valid M-Pesa payment prompt

Make Payment with M-Pesa

STK push sent successfully. Please check your phone for the payment prompt. ✕



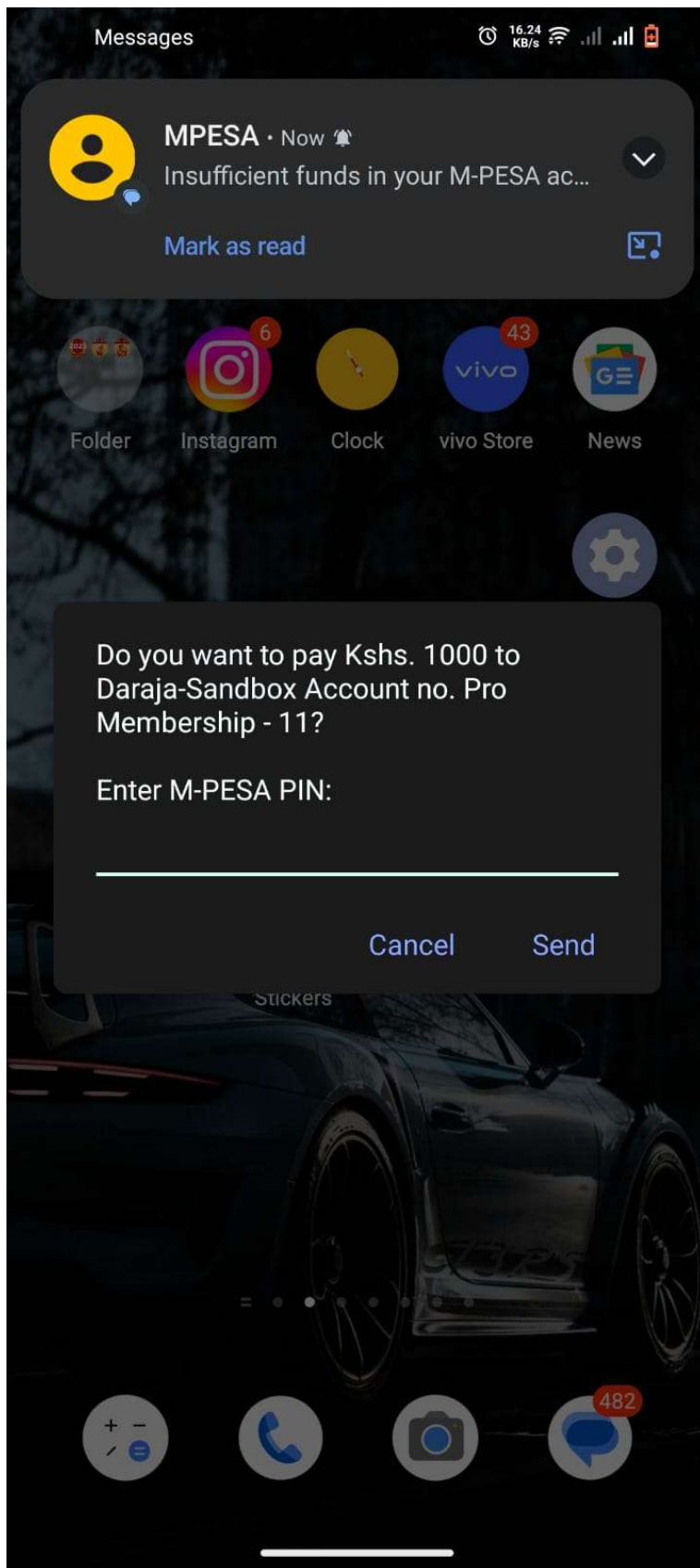
Student pro
Amount: KES 1,000
AI Prompts: 150

Phone Number (Format: 254XXXXXXXXX)

Enter your phone number in the format 254XXXXXXXXX

Amount (KES)

⌂ Processing...



Appendix 25 Invalid M-Pesa payment

Make Payment with M-Pesa



Student pro
Amount: KES 1,000
AI Prompts: 150

Phone Number (Format: 254XXXXXXXXXX)



Enter your phone number in the format 254XXXXXXXXXX

Please enter a valid phone number in the format 254XXXXXXXXXX

Amount (KES)

Pay Now

Appendix 26 M-Pesa payment prompt dismissed

Make Payment with M-Pesa

Payment failed: Prompt Dismissed



Student pro

Amount: KES 1,000

AI Prompts: 150

Phone Number (Format: 254XXXXXXXXX)

254112916170

Enter your phone number in the format 254XXXXXXXXX

Amount (KES)

1000

Try Again

Appendix 27 M-Pesa payment successful

Make Payment with M-Pesa

Payment Successfull ×



Student pro

Amount: KES 1,000

AI Prompts: 150

Phone Number (Format: 254XXXXXXXXXX)

254112916170

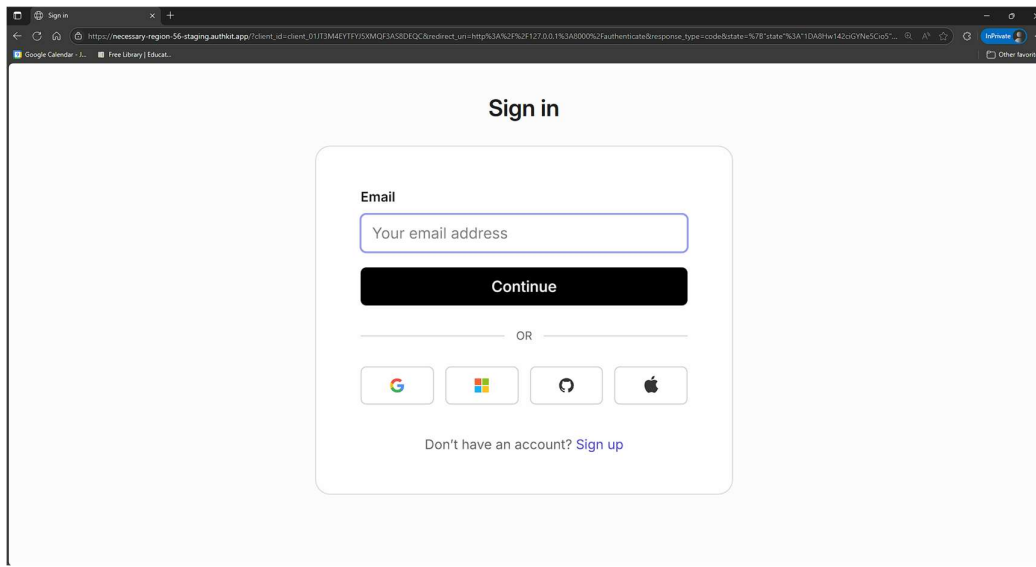
Enter your phone number in the format 254XXXXXXXXXX

Amount (KES)

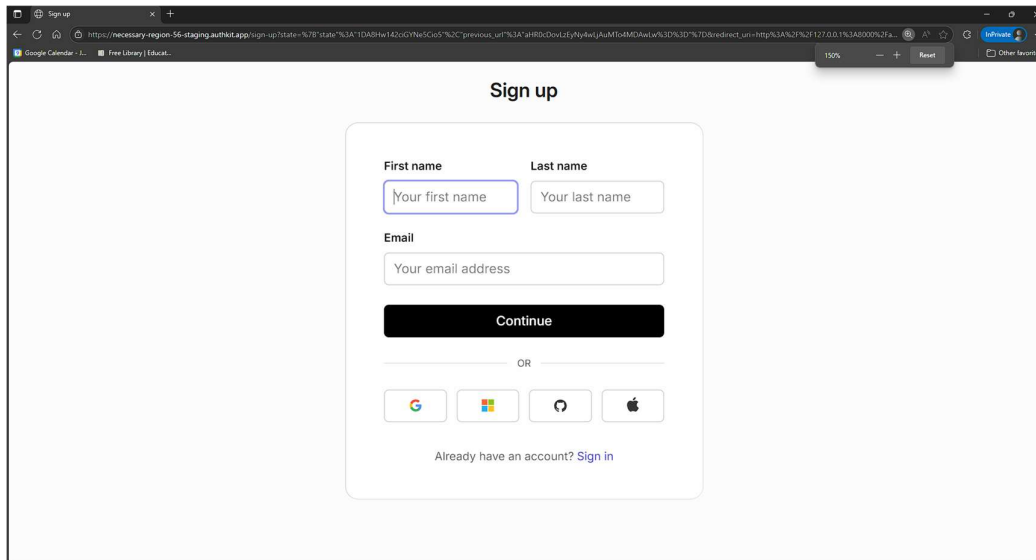
1000

back to dashboard

Appendix 28 User Authentication



A screenshot of a web browser displaying a "Sign in" page. The page has a light gray background. In the center, there is a white rounded rectangle containing the sign-in form. At the top of this rectangle, the text "Sign in" is displayed in a bold, black font. Below this, the label "Email" is positioned above a text input field with the placeholder text "Your email address". Underneath the input field is a black button with the word "Continue" in white. Below the button, the word "OR" is centered between two horizontal lines. Underneath these lines are four social media login buttons: Google, Microsoft, GitHub, and Apple. At the bottom of the white rectangle, the text "Don't have an account? [Sign up](#)" is displayed, with "Sign up" being a blue link.



A screenshot of a web browser displaying a "Sign up" page. The page has a light gray background. In the center, there is a white rounded rectangle containing the sign-up form. At the top of this rectangle, the text "Sign up" is displayed in a bold, black font. Below this, there are two input fields for "First name" and "Last name", each with placeholder text "Your first name" and "Your last name" respectively. Below these is an "Email" label above a text input field with the placeholder text "Your email address". Underneath the input fields is a black button with the word "Continue" in white. Below the button, the word "OR" is centered between two horizontal lines. Underneath these lines are four social media login buttons: Google, Microsoft, GitHub, and Apple. At the bottom of the white rectangle, the text "Already have an account? [Sign in](#)" is displayed, with "Sign in" being a blue link.

Appendix 29 AI Assisted Task Creation

Let AI help you create and distribute tasks

Create New Assignment with AI

Describe the assignment and tasks in detail. The AI will generate structured tasks based on your description and distribute them among team members.

Group: IS project | Created by: Eeshan vaghjani | less than a minute ago
mistralai/mistral-7b-instruct
Original Prompt:

You are an expert educator and task designer. Create a structured, student-friendly assignment using the following details: Topic: Academic Workflow Management Subject: Information Systems Grade/Level: Undergraduate / third Year Assignment Type: Final Project Documentation Learning Outcome/Objective: Students will demonstrate their understanding of academic workflow systems by producing comprehensive documentation that evaluates, designs, and implements key solution modules focused on automation, transparency, and usability. Chapters Covered: Chapter 4 - System Design Chapter 5 - Implementation & Testing Chapter 6 - Evaluation & Conclusion Deadline: July 11, 2025

Due: 11/07/2025

Be specific about what the assignment entails, including requirements, format, learning objectives, and any other relevant details.

Press **Ctrl+Enter** to submit

Academic Workflow Management Final Project

Information Systems

Description

Students will demonstrate their understanding of academic workflow systems by producing comprehensive documentation that evaluates, designs, and implements key solution modules focused on automation, transparency, and usability. Chapters covered: Chapter 4 - System Design, Chapter 5 - Implementation & Testing, Chapter 6 - Evaluation & Conclusion.

Due: 11/07/2025

Generated Tasks

Redistribute Tasks

Task	Assigned To	Due Date	Priority	Effort	Importance
Research and Evaluate Academic Workflow Systems Research, document, and evaluate existing academic workflow systems focusing on their strengths, weaknesses, and best practices. Propose improvements and create a design plan.	Eeshan vaghjani	06/07/2025	high	⌚ 6h ⌚ 5/5	
Design System Architecture and Data Flow Create a detailed system architecture design and data flow diagram for the proposed academic workflow system, covering all chapters.	Dhruvin Bhudia	10/07/2025	medium	⌚ 8h ⌚ 4/5	
Develop Automated Task Lists and Scheduling Develop automation features for task lists and scheduling to ensure efficient task management within the academic workflow system.	Eeshan vaghjani	13/07/2025	medium	⌚ 6h ⌚ 3/5	
Create User Guides and Manuals Develop user-friendly guides and manuals for each module in the academic workflow system, detailing its features, usage, and troubleshooting procedures.	Dhruvin Bhudia	15/07/2025	low	⌚ 4h ⌚ 2/5	
Test and Refine System Features Test all features of the academic workflow system, document any issues, and work with the design and development teams to refine and improve the system.	Eeshan vaghjani	16/07/2025	medium	⌚ 4h ⌚ 3/5	
Prepare Final Documentation and Presentations Compile all research, evaluations, design plans, and solution modules into a comprehensive final documentation and create presentations detailing the project's progress and findings.	Dhruvin Bhudia	22/07/2025	high	⌚ 7h ⌚ 5/5	

Member Workload Preview

This shows how workload would be distributed if these tasks are saved.

Note: Tasks are not saved until you click "Save Assignment"

E Eeshan vaghjani
3 tasks

38.5%
Total Effort: 10 hours

Task Details:
Total Effort: 10 hours
Importance Score: 3

Chapter 4 System Design Research Chapter 5 Implementation & Testing Research Chapter 6 Evaluation & Conclusion Research

D Dhruvin Bhudia
3 tasks

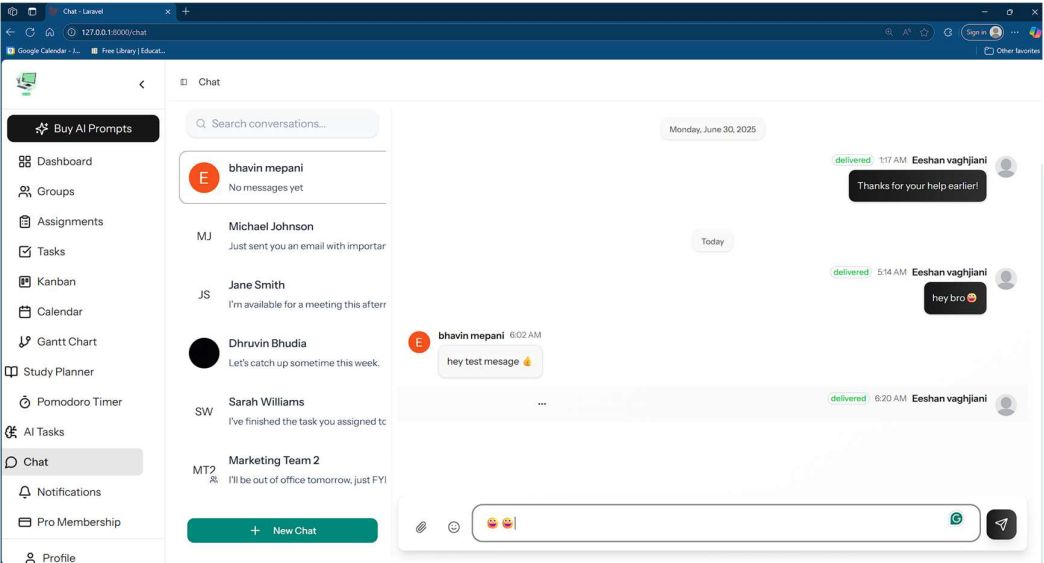
61.5%
Total Effort: 16 hours

Task Details:
Total Effort: 16 hours
Importance Score: 3

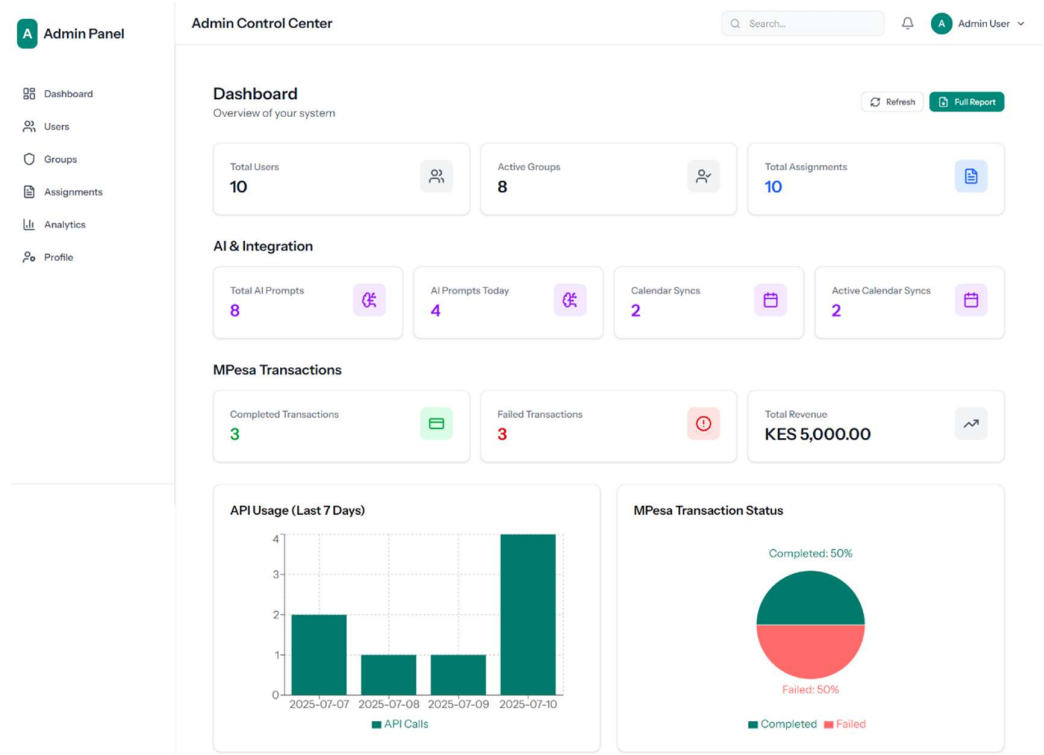
Chapter 4 System Design Implementation Chapter 5 Implementation & Testing Implementation Chapter 6 Evaluation & Conclusion Implementation

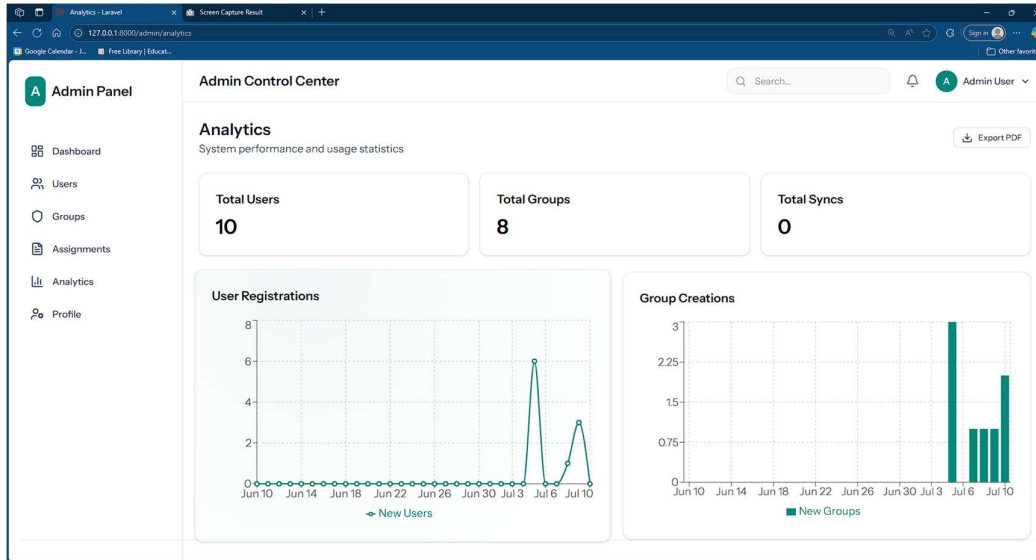
Save Assignment

Appendix 30 Chat System



Appendix 31 Admin dashboard





Admin Panel

Admin Control Center

User Management
Manage user accounts and permissions

Search users by name or email...

Filter Export

Name	Email	Role	Created	Last Login	Status	Actions
simpleave	simpleave33@gmail.com	USER	July 9th, 2025	Never	Deleted	
dhravin bhudia	dhravin.bhudia@strathmore.edu	ADMIN	July 9th, 2025	July 10th, 2025	Active	
bhavin mepani	evaghjani04@gmail.com	USER	July 9th, 2025	July 10th, 2025	Active	
Krishna Mahendra	krishnamahendra2004@gmail.com	USER	July 8th, 2025	Never	Active	
Sarah Williams	sarah@example.com	USER	July 5th, 2025	Never	Active	
Dhravin Bhudia	dhravinbhudiael11@gmail.com	ADMIN	July 5th, 2025	July 9th, 2025	Active	

Admin Panel

Admin Control Center

Manage Groups
A list of all groups in the system.

Refresh Export PDF

Search by name or owner...

Name	Owner	Members	Status	Created At	Actions
Microprocessor	Eeshan vaghjani	2	Active	July 10th, 2025	
HCI	dhravin bhudia	2	Active	July 9th, 2025	
OR	Krishna Mahendra	2	Active	July 8th, 2025	
Final Project Team	Dhravin Bhudia	2	Active	July 7th, 2025	
Marketing Team 2	Eeshan vaghjani	3	Active	July 5th, 2025	
Development Team	Eeshan vaghjani	2	Active	July 5th, 2025	

Admin Panel

Dashboard

Users

Groups

Assignments

Analytics

Profile

Admin Control Center

Search...

Admin User

HCI

Group Details

Back to Groups

Edit

Delete

Group Information

Name

HCI

Description

hci group work

Owner

dhrvin.bhudia (dhrvin.bhudia@strathmore.edu)

Visibility

Private

Created

July 9th, 2025

Members

2 members

Members

Name	Email	Joined
Eeshan vaghiani	evaghiani@gmail.com	N/A
dhrvin bhudia	dhrvin.bhudia@strathmore.edu	N/A

Assignments

Title	Due Date	Tasks	Created
HCIschool/Website Mid-Fidelity Prototype	Jul 13, 2025		Jul 9, 2025

Admin Panel

Dashboard

Users

Groups

Assignments

Analytics

Profile

Admin Control Center

Search...

Admin User

Assignments Management

Export PDF

Search Assignments

Search by title, description or

Search

Title	Group	Created By	Due Date	Tasks	Status	Actions
HCIschool/Website Mid-Fidelity Prototype	HCI	N/A	13/07/2025 Due soon	4	Active	
Optimum Advertisement Allocation	OR	N/A	15/07/2025 Upcoming	4	Active	
Capstone Project Design	Development Team	N/A	01/12/2025 Upcoming	6	Active	
Website Redesign	Marketing Team 2	N/A	18/07/2025 Upcoming	4	Active	
Website Redesign	Development Team	N/A	30/07/2025 Upcoming	6	Active	
Documentation Update	Development Team	N/A	24/07/2025 Upcoming	5	Active	
Marketing Campaign	Development Team	N/A	22/07/2025 Upcoming	4	Active	
Website Redesign	N/A	N/A	24/07/2025 Upcoming	6	Active	
Marketing Campaign	Client Project	N/A	01/08/2025 Upcoming	4	Active	
Documentation Update	Client Project	N/A	26/07/2025 Upcoming	3	Active	

92

Admin Panel

Dashboard

Users

Groups

Assignments

Analytics

Profile

Admin Control Center

Search...

Admin User

HCIschoolWebsite Mid-Fidelity Prototype

Edit

Group

Created By

Due Date

HCI

N/A

July 13, 2025

Description

Details

Create a mid-fidelity prototype of a website aimed at school students, submit as part of HCI assignment due on Monday

Unit Name
HCI Assignment
Created At
July 9, 2025
Last Updated
July 9, 2025

Tasks (4)

0 of 4 completed 0%

☒ Wireframe Homepage and About Us Page

Design wireframes for the homepage and about us page of the HCIschoolWebsite prototype

☒ Create Navigation Flow and Link Pages

Create navigation flow and link all pages using a free prototyping tool (Figma or Framer) for the HCIschoolWebsite prototype

☒ Final Review

Review and provide feedback on all aspects of the mid-fidelity prototype of the HCIschoolWebsite

☒ Design Student Login and Registration Pages

Design login and registration pages for the HCIschoolWebsite prototype

Appendix 32 Reports and Analytics

User Management Report

Generated on: 10/07/2025, 09:41:34

Active Users

ID	Name	Email	Role	Created	Last Login
10	dhrvin bhudia	dhrvin.bhudia@strathmore.edu	ADMIN	July 9th, 2025	July 10th, 2025
9	bhavin mepani	evaghjiani04@gmail.com	USER	July 9th, 2025	July 10th, 2025
8	Krishna Mahendra	krishnamahendra2004@gmail.com	USER	July 8th, 2025	Never
5	Sarah Williams	sarah@example.com	USER	July 5th, 2025	Never
6	Dhrvin Bhudia	dhrvinbhudiael11@gmail.com	ADMIN	July 5th, 2025	July 9th, 2025
7	Admin User	admin@example.com	ADMIN	July 5th, 2025	Never
1	Eeshan vaghjiani	evaghjiani@gmail.com	ADMIN	July 5th, 2025	July 10th, 2025
3	Jane Smith	jane@example.com	USER	July 5th, 2025	Never

Deleted Users

ID	Name	Email	Role	Created	Deleted At
11	simpleave	simpleave33@gmail.com	USER	July 9th, 2025	July 10th, 2025
2	John Doe	john@exaple.com	USER	July 5th, 2025	July 9th, 2025

Analytics Report

Generated on July 10, 2025

Key Metrics

Total Users
9

Active Users (Last 7 Days)
4

Total Groups
7

Active Groups (Last 7 Days)
7

User Activity Summary

Metric	Count	Percentage
Active Users	4	44.4%
Inactive Users	5	55.6%
Active Groups	7	100%
Inactive Groups	0	0%

This report is system generated. For any queries, please contact the administrator.